

Aim

Conclusions

1. Does it work, and is it consistent
2. Distribution across cell types
3. Biotin-high versus biotin-low in HSC/MPP
- 3b. The same split in monocytes
4. What was run, in order
5. Data locations
6. Tools
7. Caveats

Biotin report

Code ▼

2026-08-05

Samples: biotin was compared across LK1 and LK2, and the high/low split was done on WEI21_26-17_NK from LK3.

Status: the labelling works. The high-versus-low split is exploratory.

Analysis finished: 2026-08-05 - the most recent of the 49 figures shown below.

Report generated: 2026-08-05 by 04. biotin_report.Rmd , from the outputs of steps 01-03 in this folder.

Biotin is carried on the ADT panel as a readout of cell-surface glycosylation - “sugar-seq”. It was added deliberately from the pilot run onward, so every CITE-seq experiment carries it alongside the antibody panel. This report covers how much biotin signal there is, how it is distributed, and what separates biotin-high from biotin-low cells.

For full report, download biotin_report.html in the attachment and open in a browser.

Code used - attached and in the github repository: https://github.com/nbartonicek/surface_proteome_dev/tree/main/lab_archives/6.Biotin_report (https://github.com/nbartonicek/surface_proteome_dev/tree/main/lab_archives/6.Biotin_report)

Aim

- does the biotin labelling work at all, and is it consistent between runs
- how is biotin distributed across cell types
- what distinguishes biotin-high from biotin-low cells within one population

Conclusions

- **sugar-seq worked** - biotin signal is present and structured in every sample, not uniform noise
- **biotin is strongly cell-type dependent.** Monocytes carry markedly the most, with pro-monocytes and cDC next; lymphoid populations and erythroid precursors carry the least. The pattern holds in both normal donors and the AML sample
- the signal is spatially coherent on the projected UMAP rather than scattered, which is what rules out its being a technical artefact of the ADT panel
- within HSC/MPP, splitting on biotin gives differences in RNA, in surface protein and in cell-cycle phase, but this arm is exploratory - it has been run on one sample only
- the same split in monocytes runs but on only 160 cells, and does not separate cycling from non-cycling: both groups are ~90% G1
- glycosylation has to be read against total surface load: step 02 computes CITE residuals adjusted for that, because otherwise a cell with more surface protein of any kind looks more glycosylated

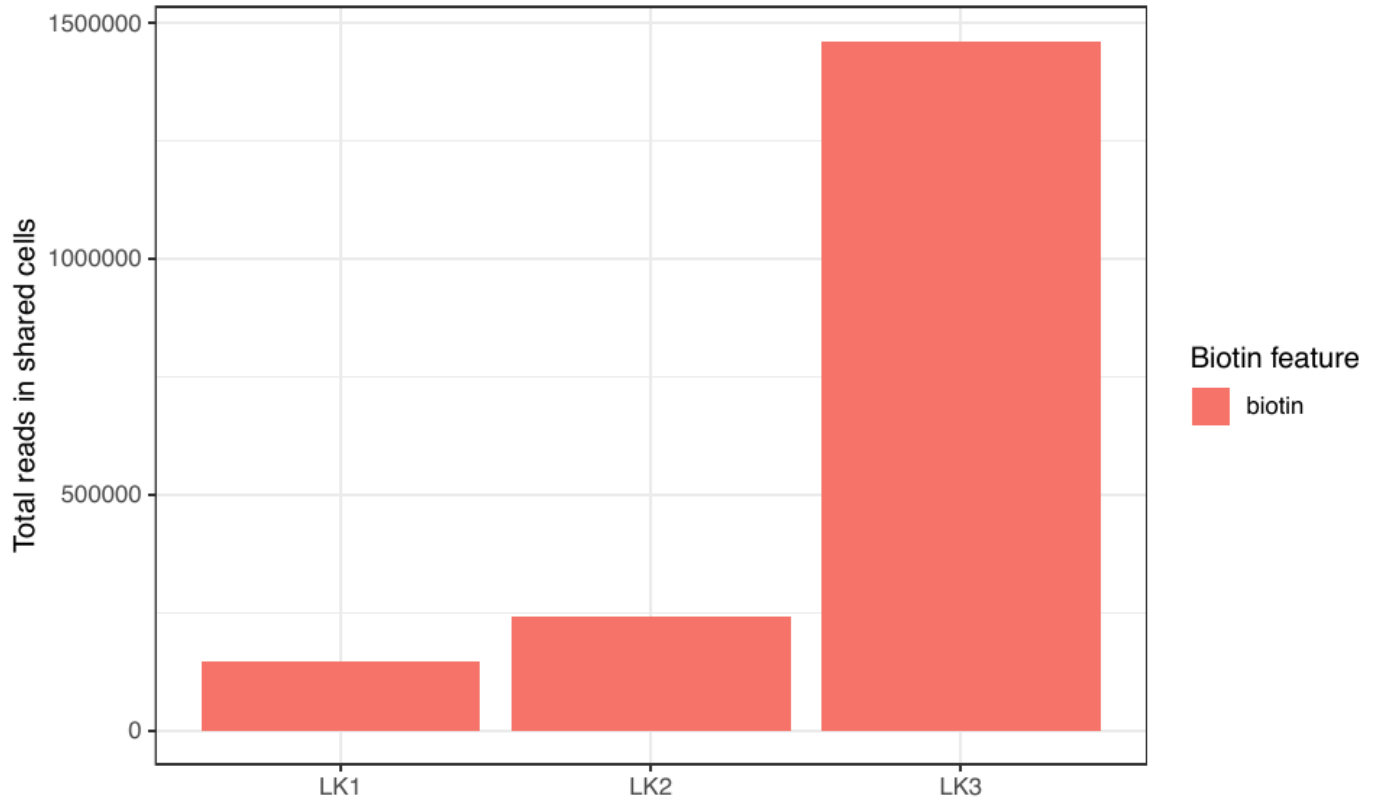
1. Does it work, and is it consistent

Step 01 compares biotin ADT levels between the LK1 and LK2 runs. It was split out of the broader HTO/ADT comparison so it can be run without the demultiplexing steps.

test_biotin_reads_before_after.pdf

Biotin ADT compared across the LK1 and LK2 runs.

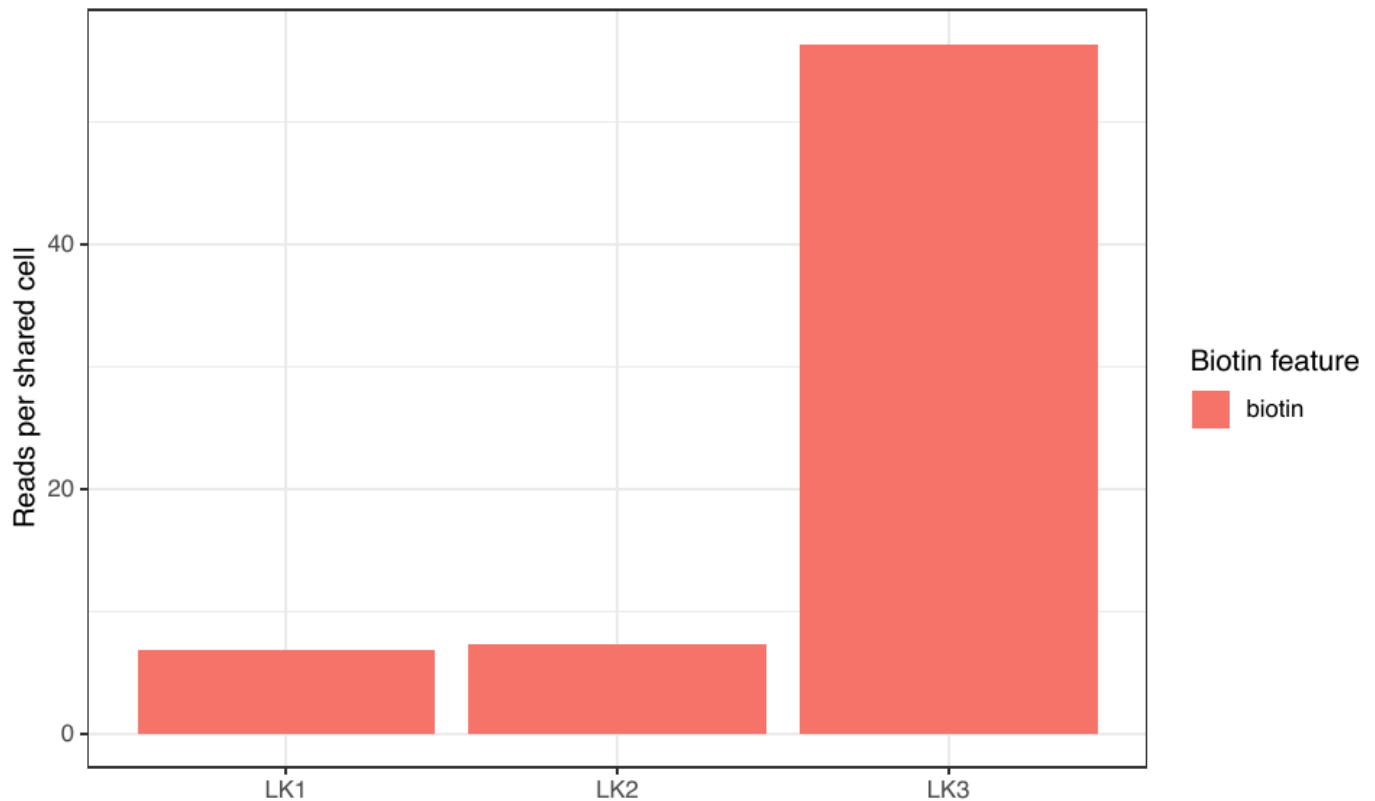
Biotin ADT reads before vs after



test_biotin_reads_per_cell_before_after.pdf

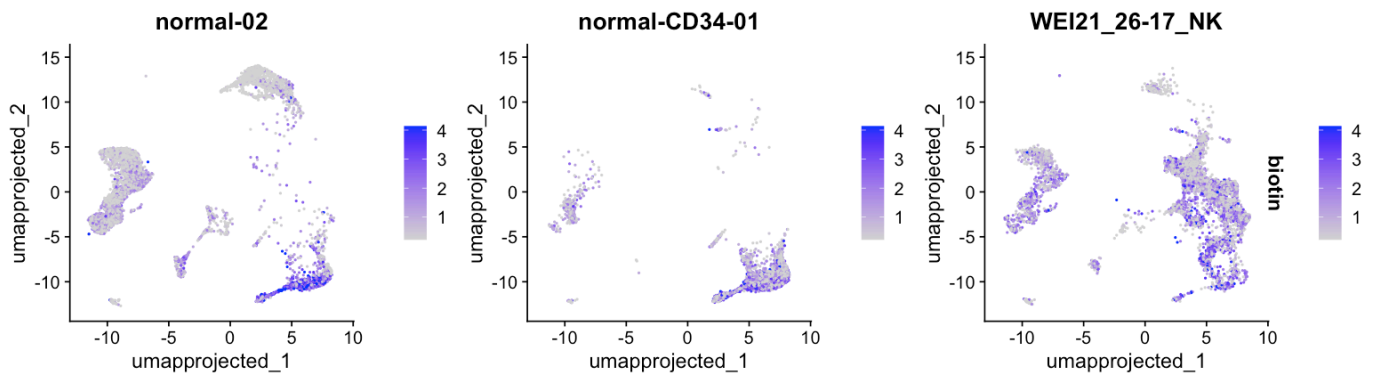
Biotin ADT compared across the LK1 and LK2 runs.

Biotin ADT reads per cell before vs after



biotin_umap_per_sample.png

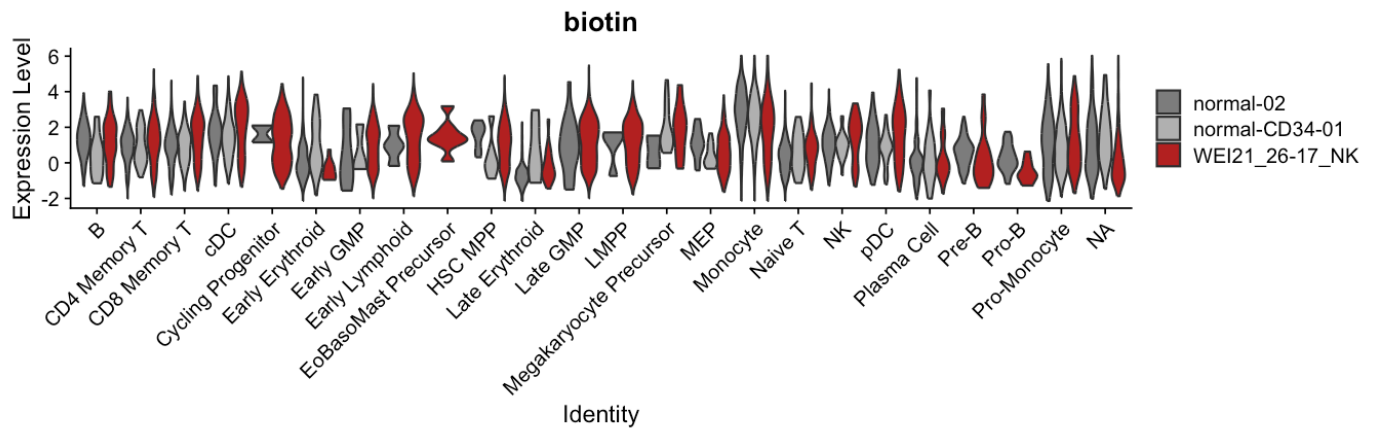
Biotin on the projected UMAP for the two normal donors and the AML sample. The signal concentrates in the same region of the map in all three, which is the evidence that the labelling worked - a technical artefact would not localise consistently to one part of the manifold across independent samples.



2. Distribution across cell types

biotin_by_celltype_violin.png

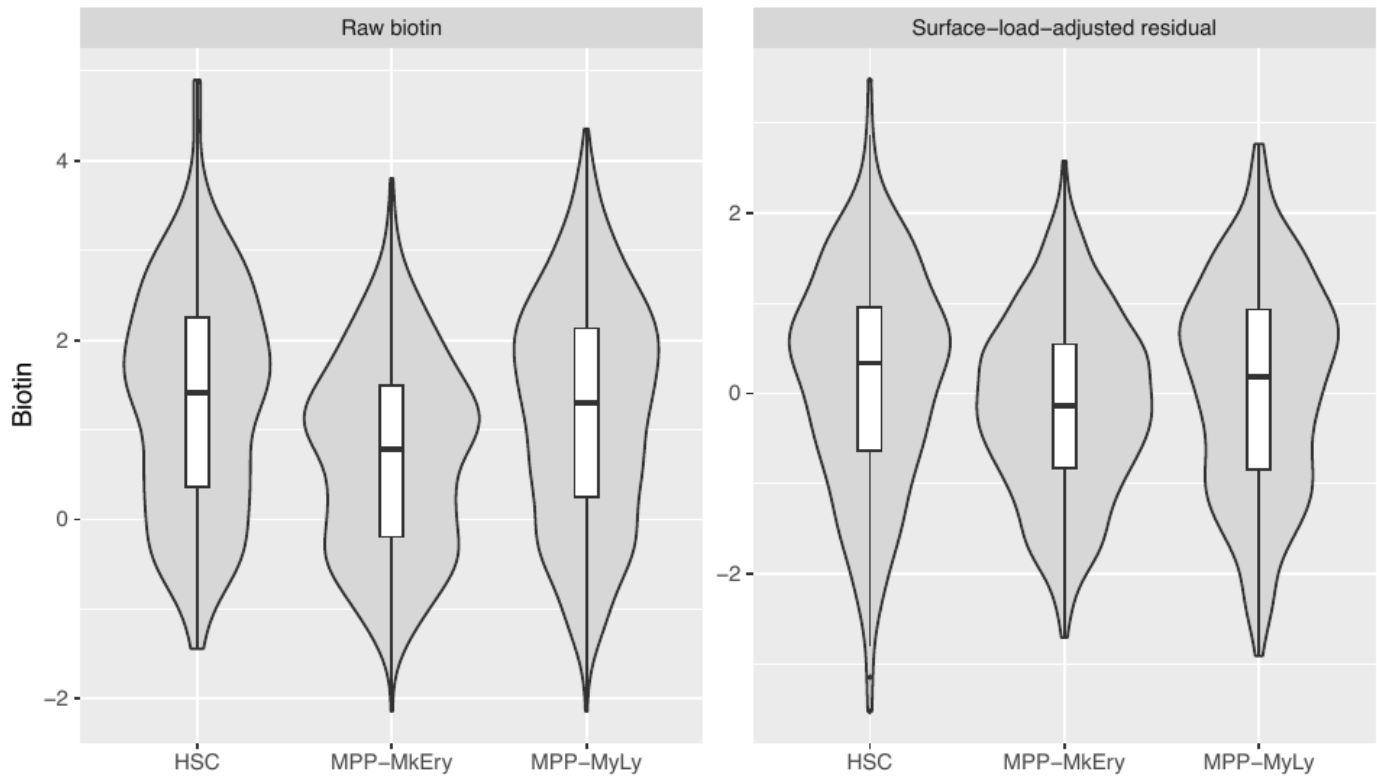
Biotin by broad cell type, split by sample. Monocytes are the clear outlier, with pro-monocytes and cDC behind them; Pro-B, Pre-B and late erythroid sit lowest. The ordering is the same in normal-02, normal-CD34-01 and WEI21_26-17_NK, so it tracks cell identity rather than sample or disease state.



3c_glyco_by_fine_celltype_violin.pdf

The same picture at the fine cell type level, from step 02.

HSC MPP (WEI21_26–17_NK) – glycosylation (biotin) by fine cell type



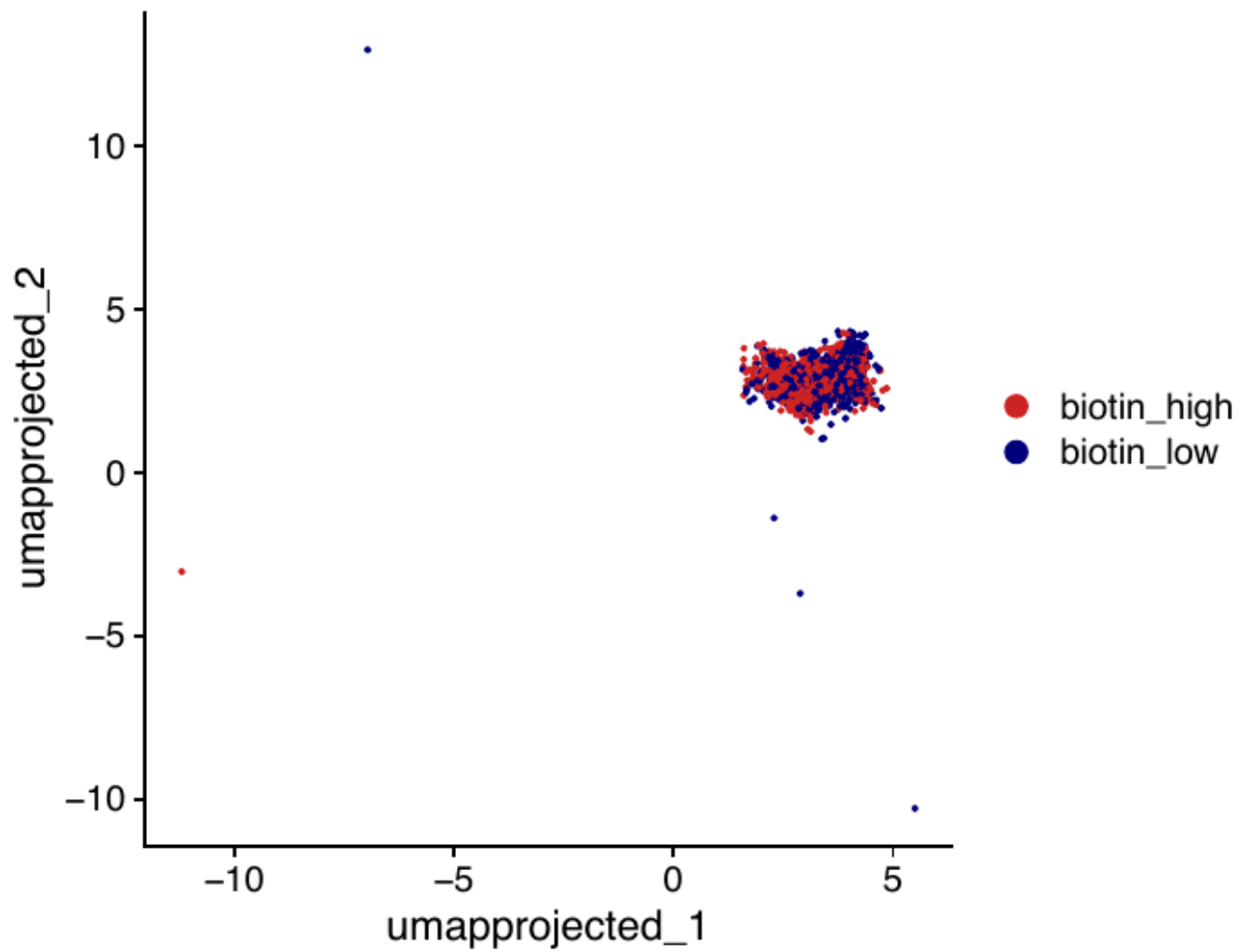
3. Biotin-high versus biotin-low in HSC/MPP

Step 02 takes the HSC/MPP compartment, splits it on CITE_DSB biotin, and asks what else differs. Everything in this section is one sample, WEI21_26–17_NK .

5a_biotin_group_umap_projected.pdf

The two groups on the projected UMAP.

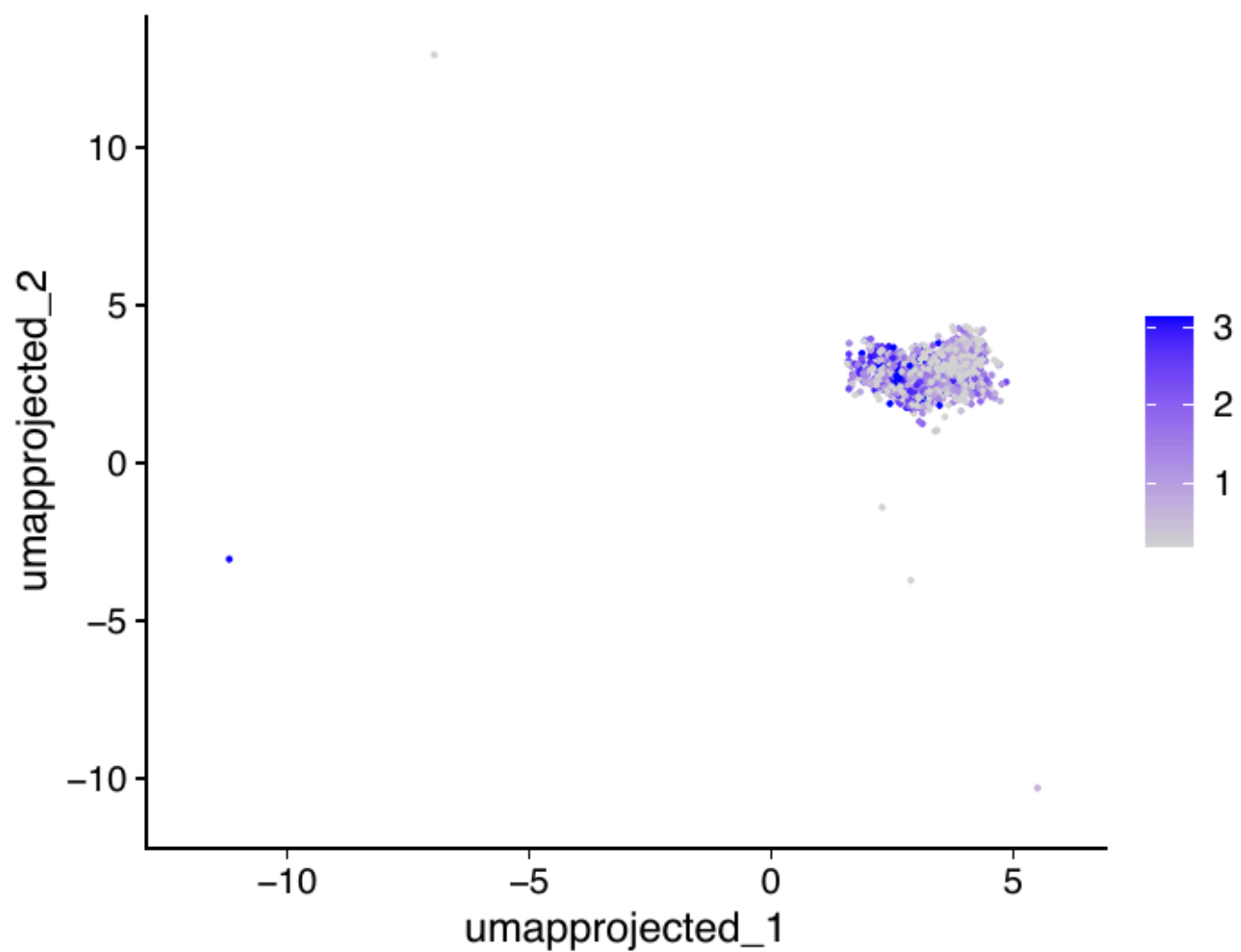
3 MPP – biotin high/low (Projected (BoneMarrowMap))



5b_biotin_value_umap_projected.pdf

Biotin as a continuous value on the same layout - the split is a threshold on this.

HSC MPP – biotin value (Projected (BoneMarrowMap))

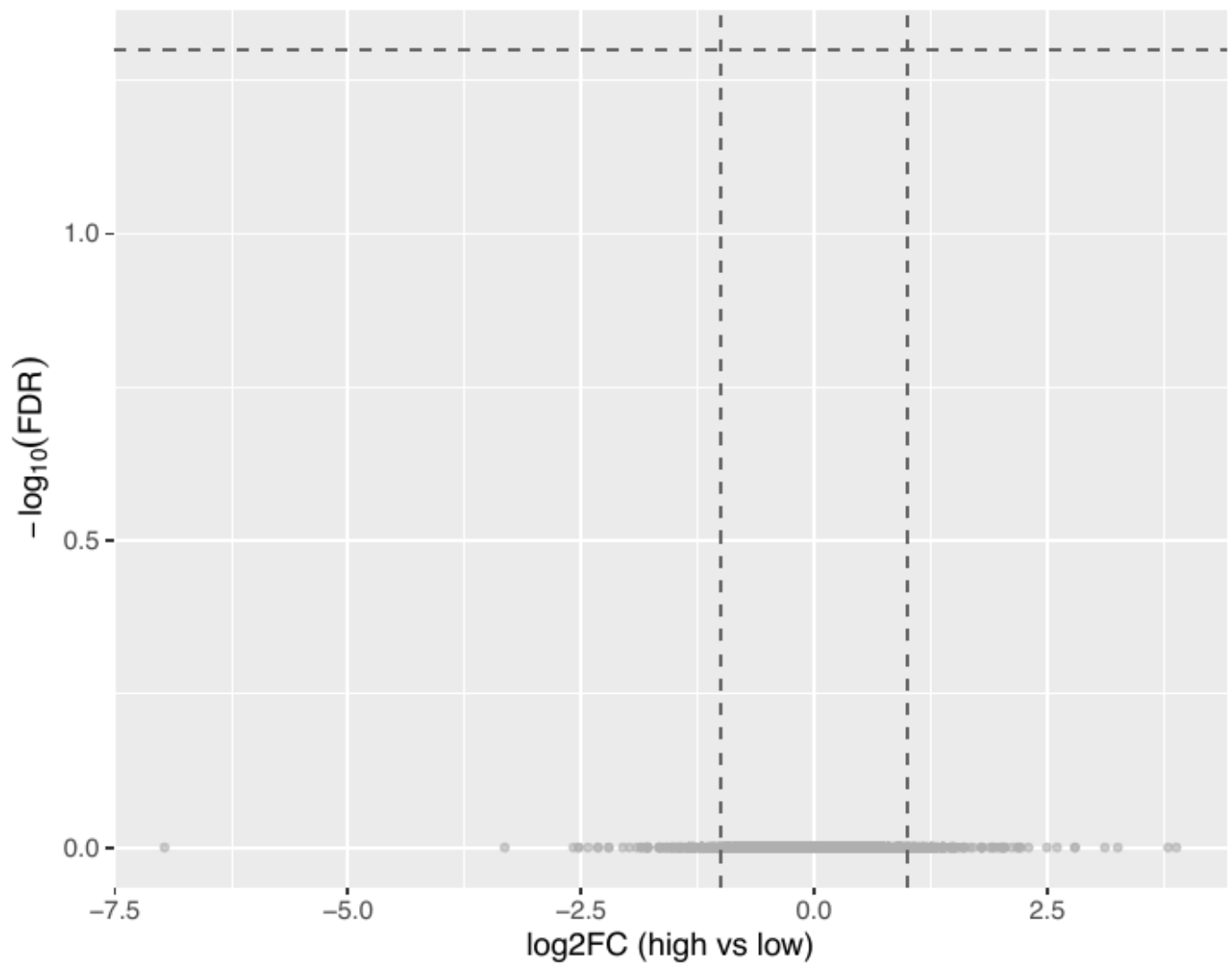


RNA

2_pseudobulk_deseq2_volcano_biotin_high_vs_low.pdf

Pseudobulk DESeq2, biotin-high against biotin-low.

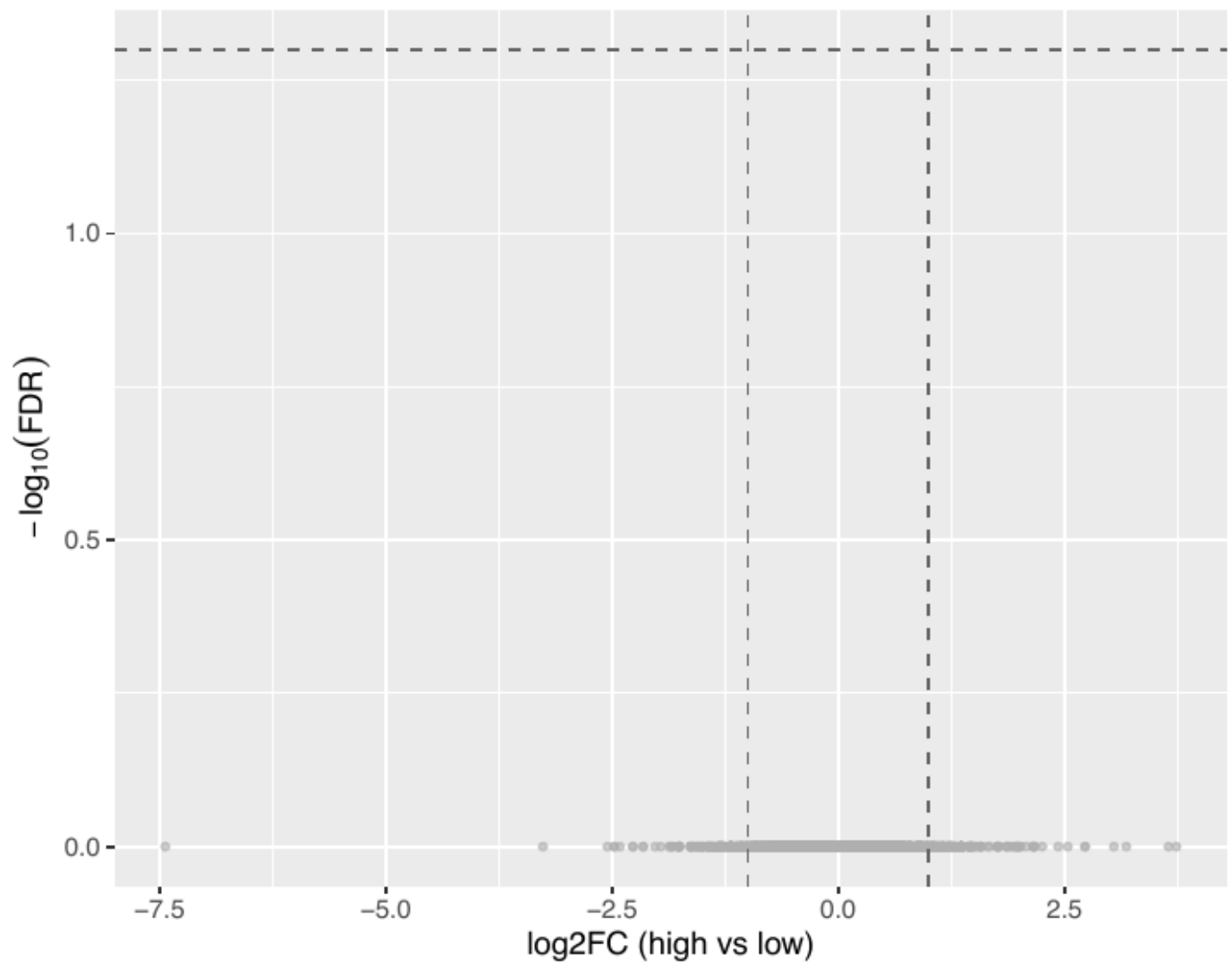
HSC MPP – pseudobulk DESeq2 volcano, biotin high vs low



2_pseudobulk_edgeR_volcano_biotin_high_vs_low.pdf

The same contrast under edgeR - shown alongside because with one sample the choice of method matters more than usual.

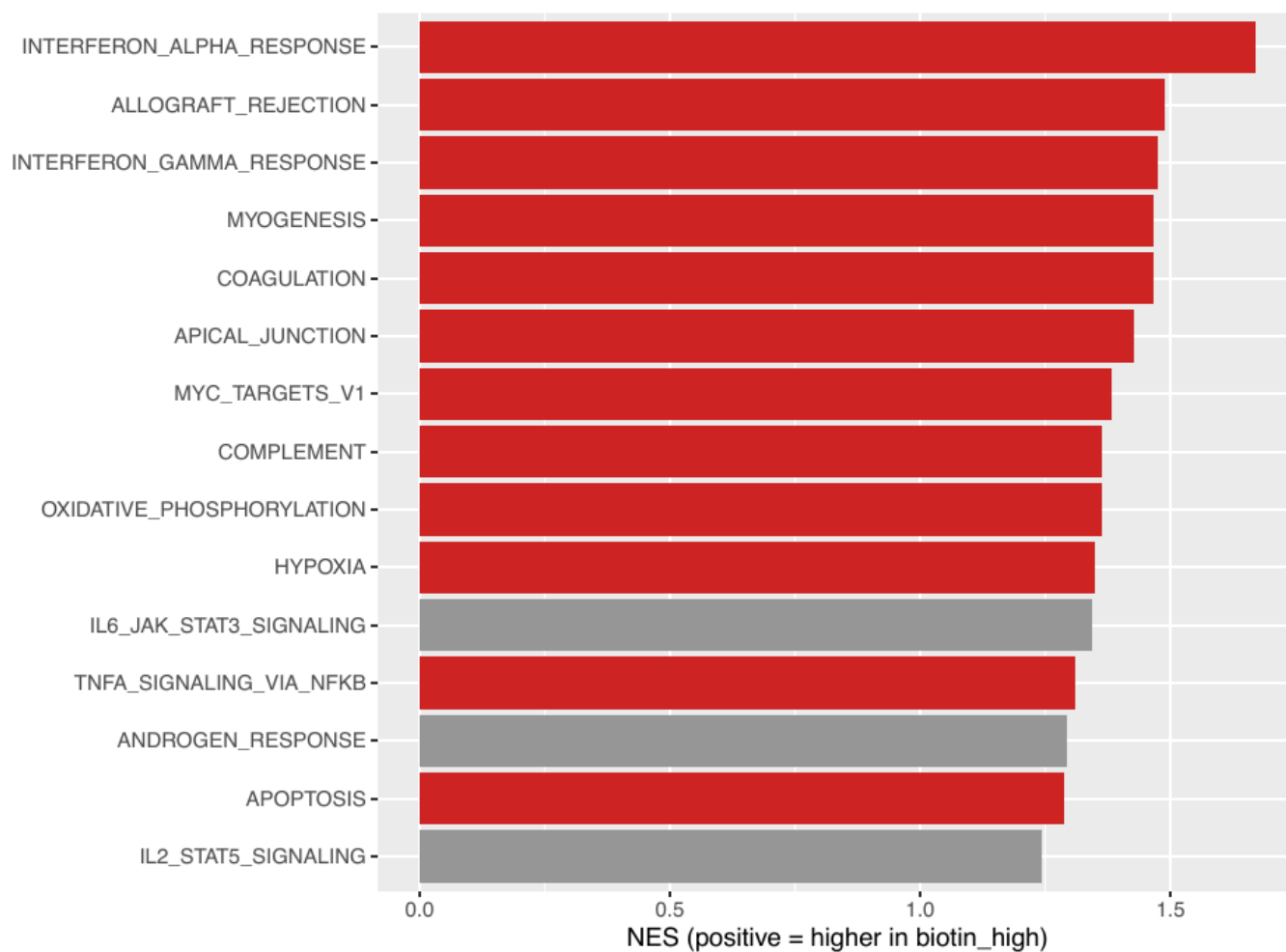
HSC MPP – pseudobulk DE volcano, biotin high vs low



1b_fgsea_hallmark_biotin_high_vs_low.pdf

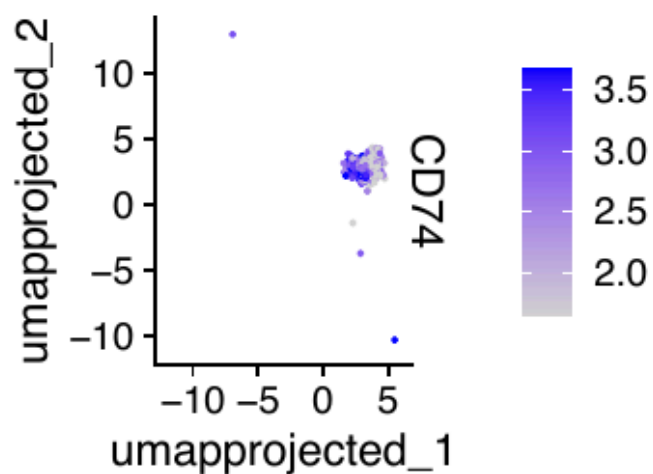
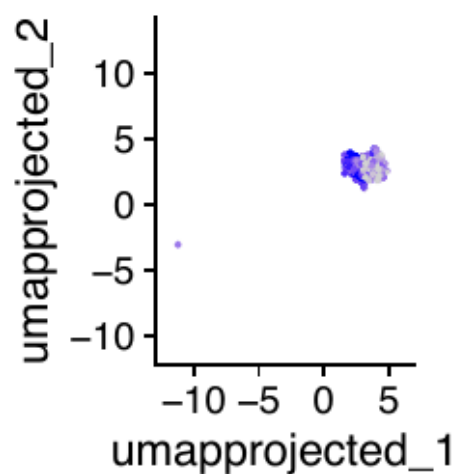
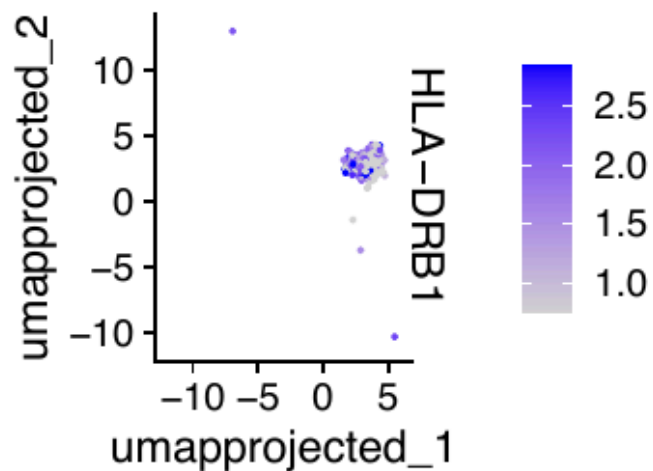
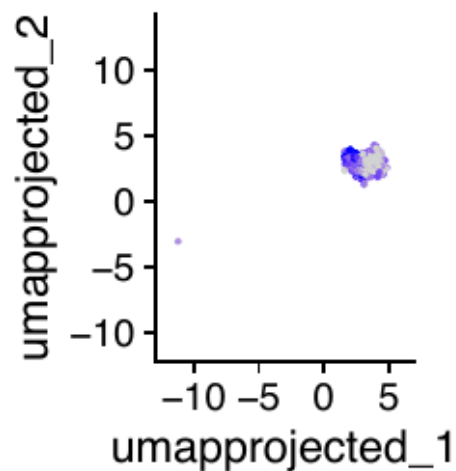
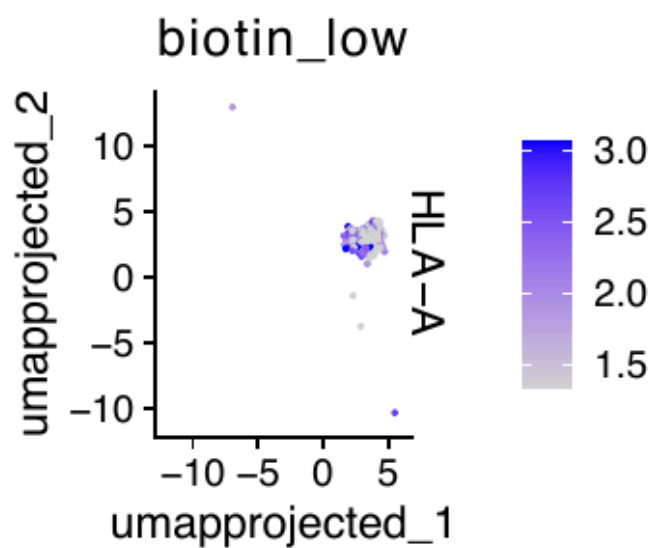
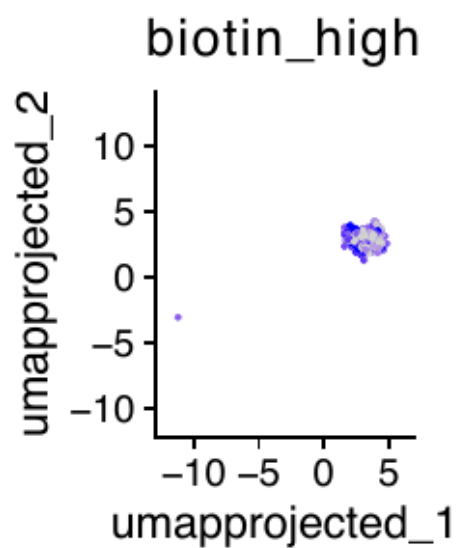
fgsea over MSigDB Hallmark, on the FindMarkers ranking.

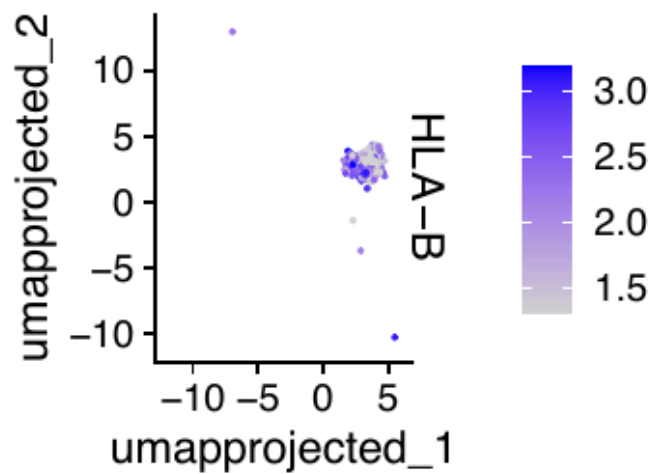
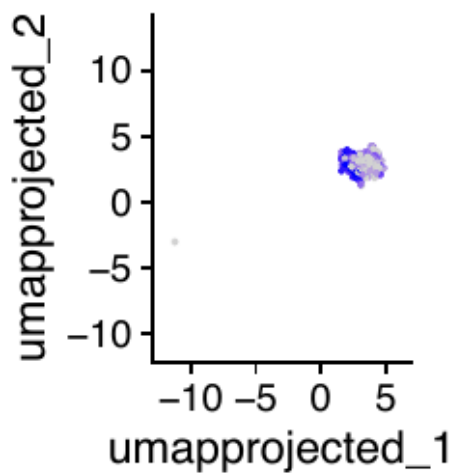
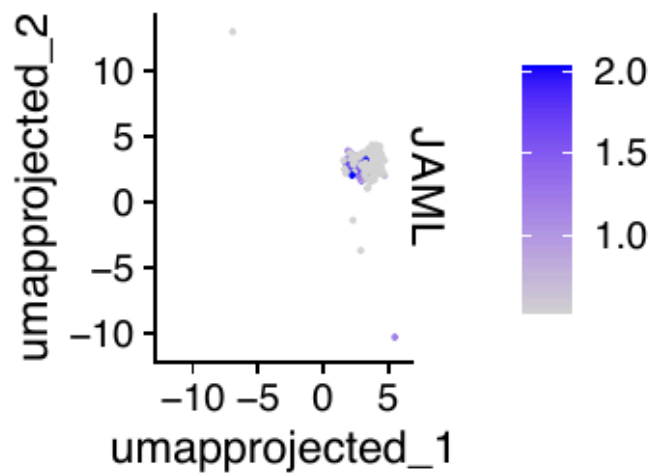
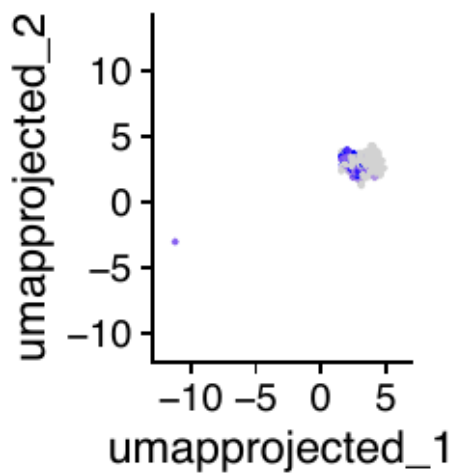
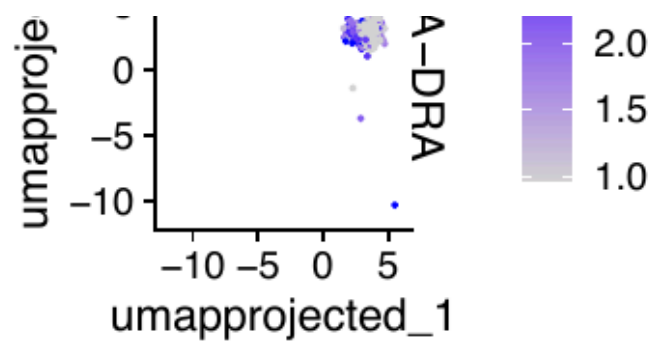
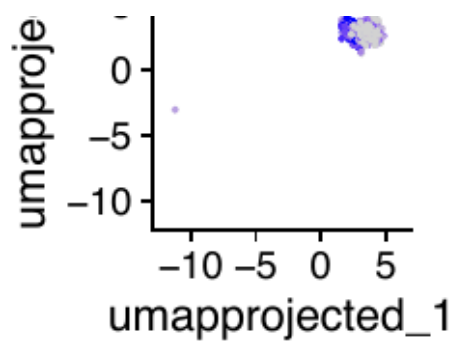
HSC MPP – top Hallmark pathways (fgsea), biotin high vs low



5d_findmarkers_top_genes_umap_projected.pdf

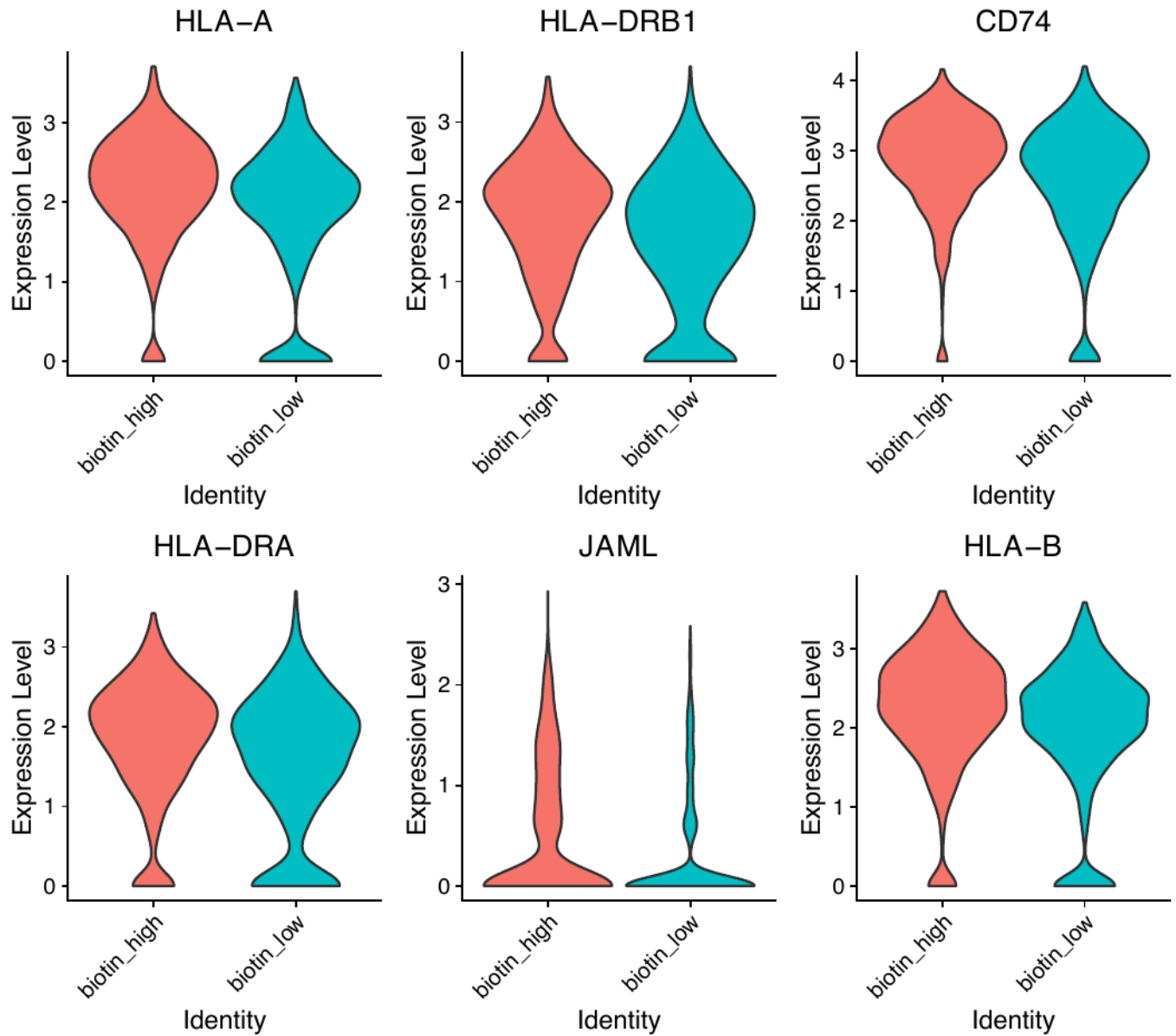
Top FindMarkers genes on the projected UMAP.





6a_findmarkers_top_genes_violin.pdf

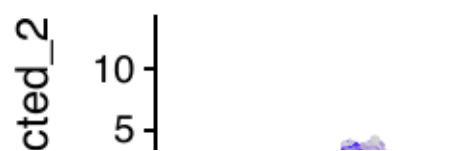
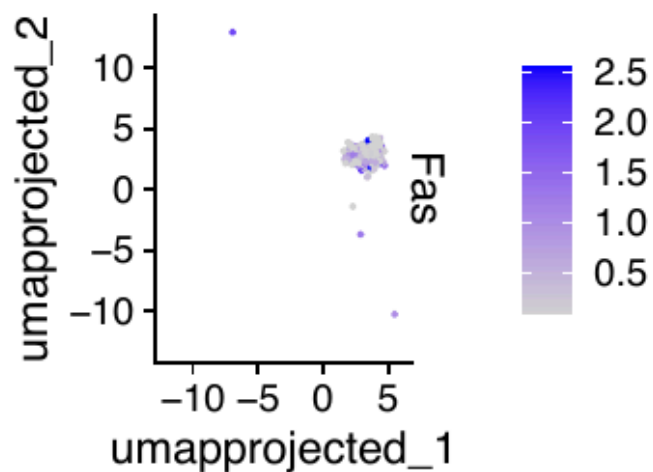
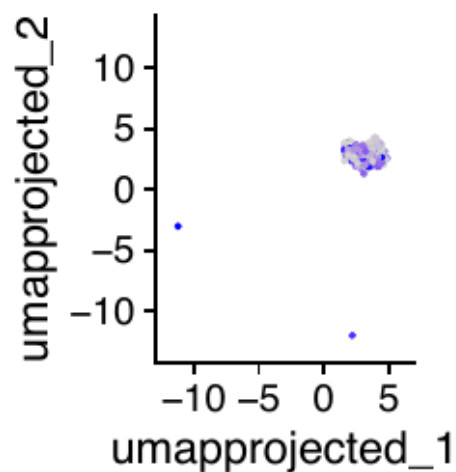
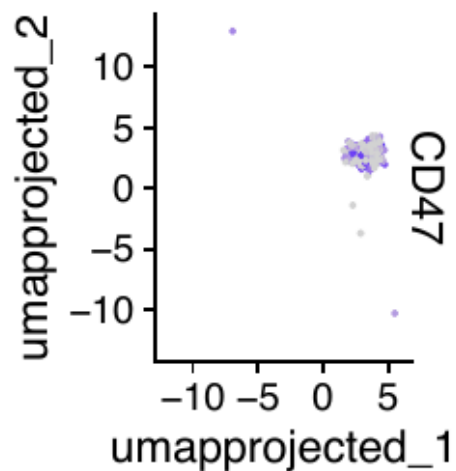
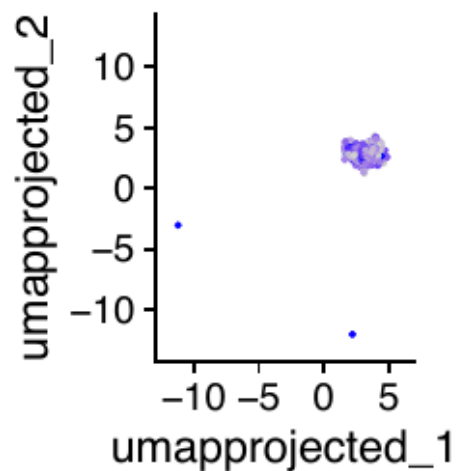
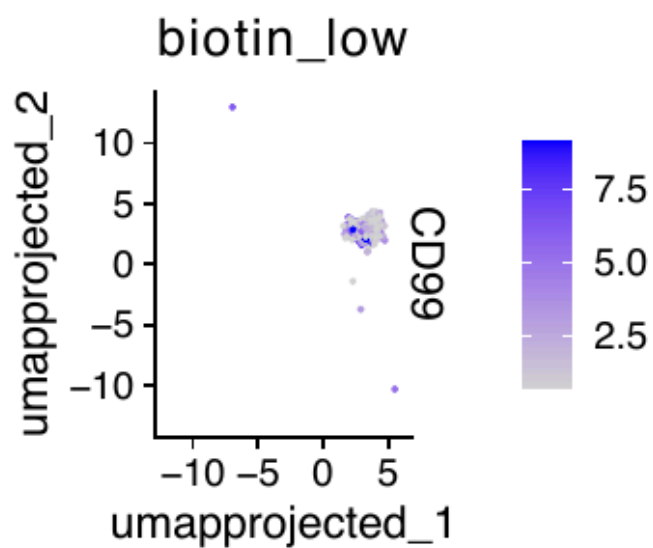
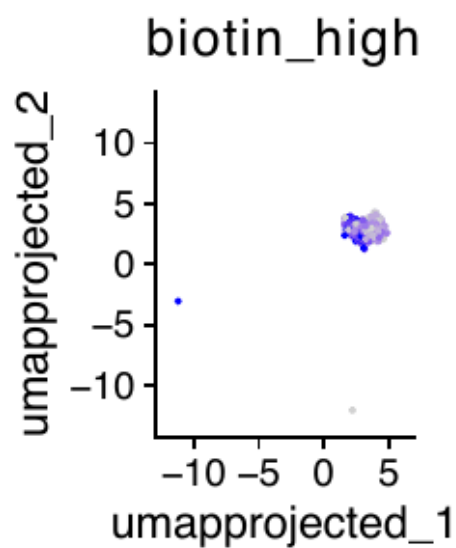
The same genes as violins by group.

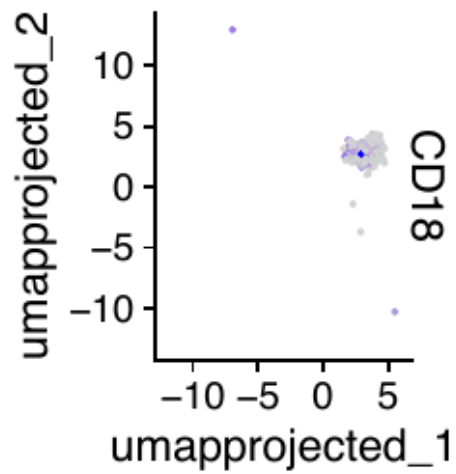
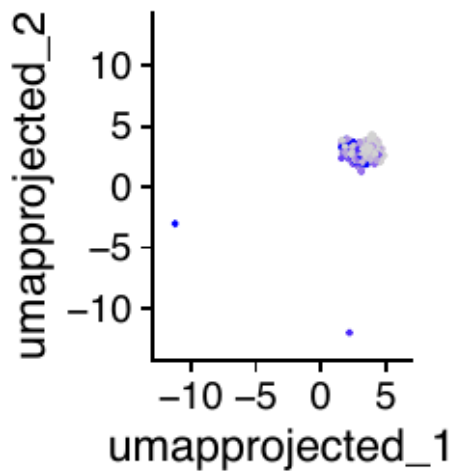
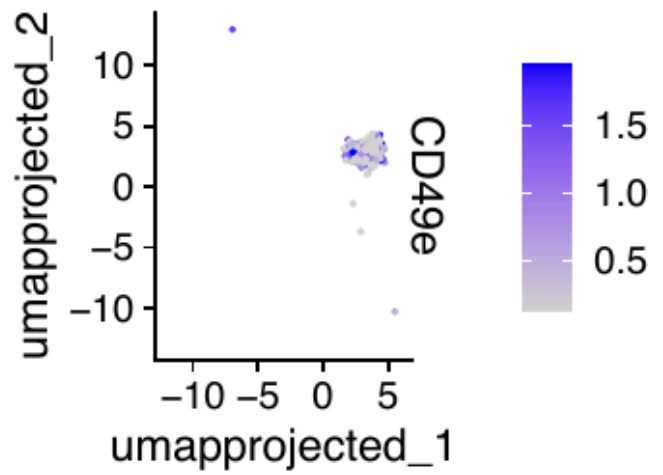
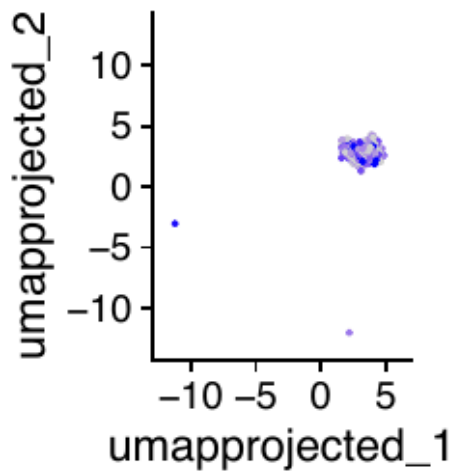
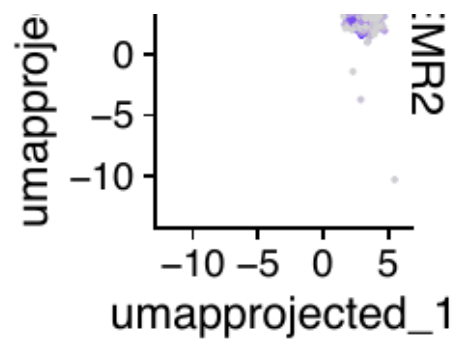
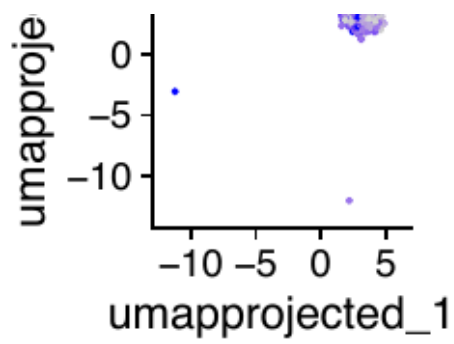


Surface protein

5f_cite_dsb_top_proteins_umap_projected.pdf

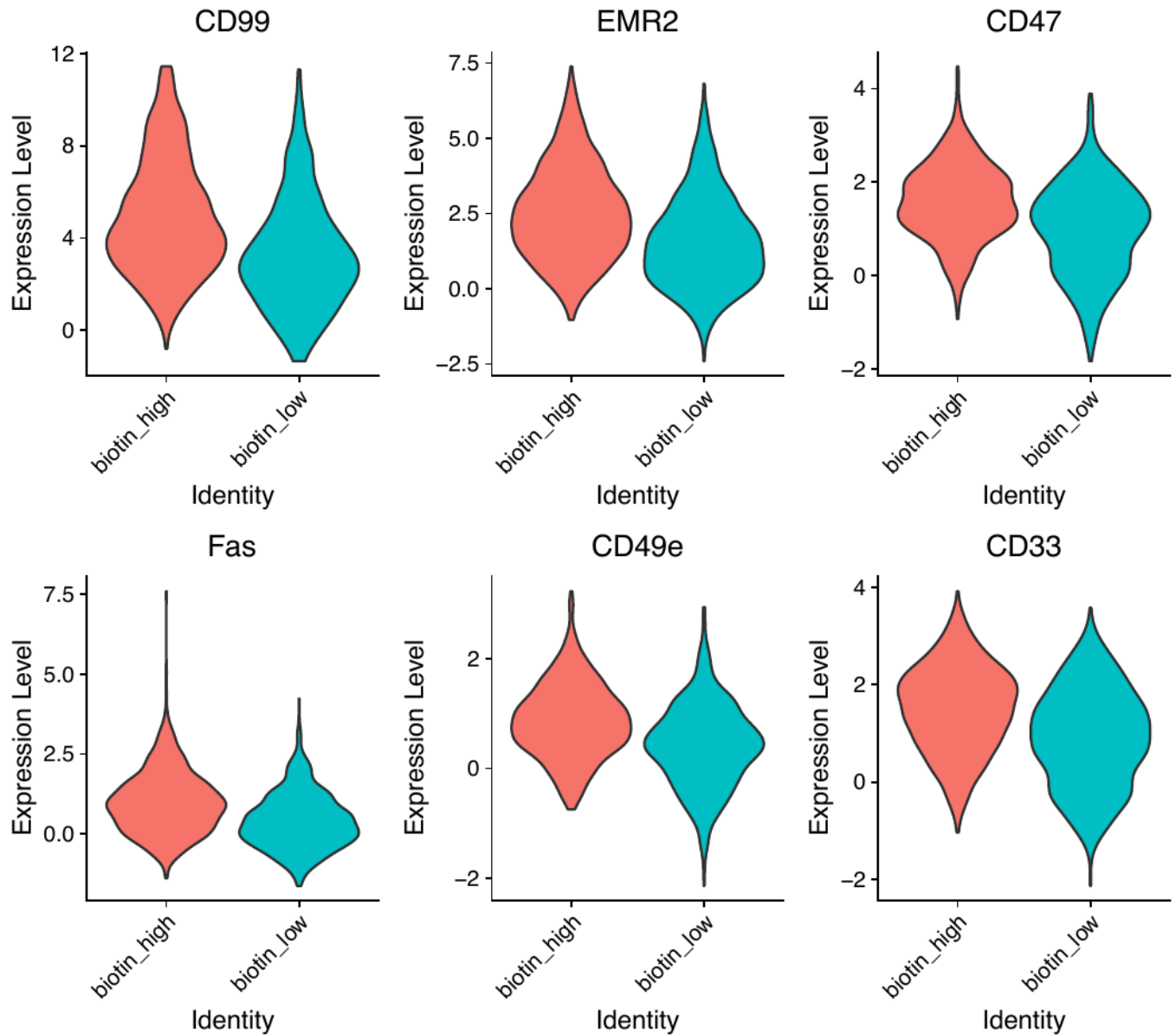
Top differential CITE_DSB proteins on the projected UMAP.





6c_cite_dsb_top_proteins_violin.pdf

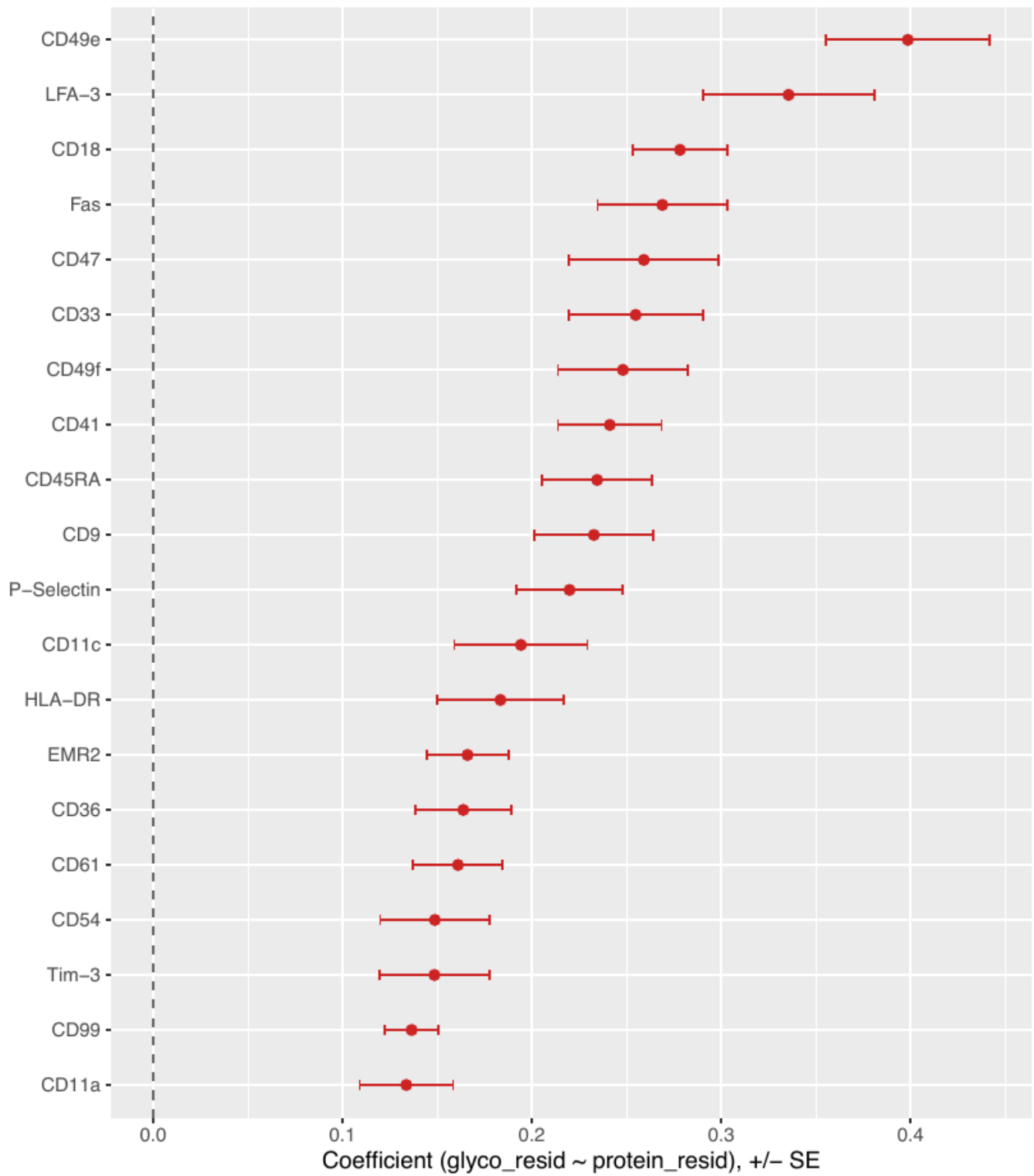
The same proteins as violins by group.



3b_glyco_residual_vs_protein_summary.pdf

Glycosylation against total surface protein, as residuals. This is the control that matters: without adjusting for surface load, any cell carrying more protein of any kind would appear more glycosylated.

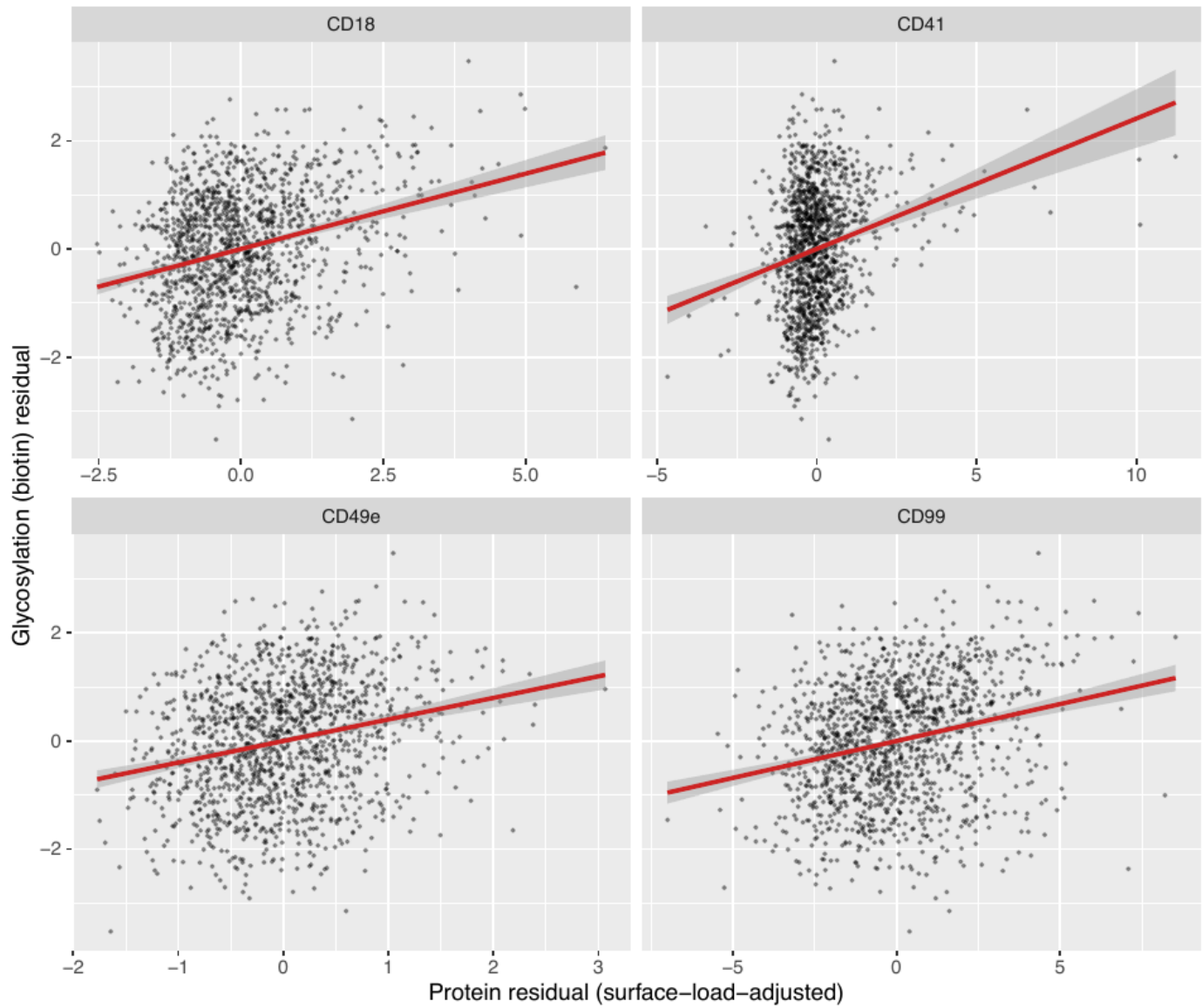
HSC MPP (WEI21_26-17_NK) – proteins vs glycosylation residual



3b_glyco_residual_vs_protein_top_scatter.pdf

The individual proteins behind that summary.

HSC MPP (WEI21_26–17_NK) – top proteins vs glycosylation residual

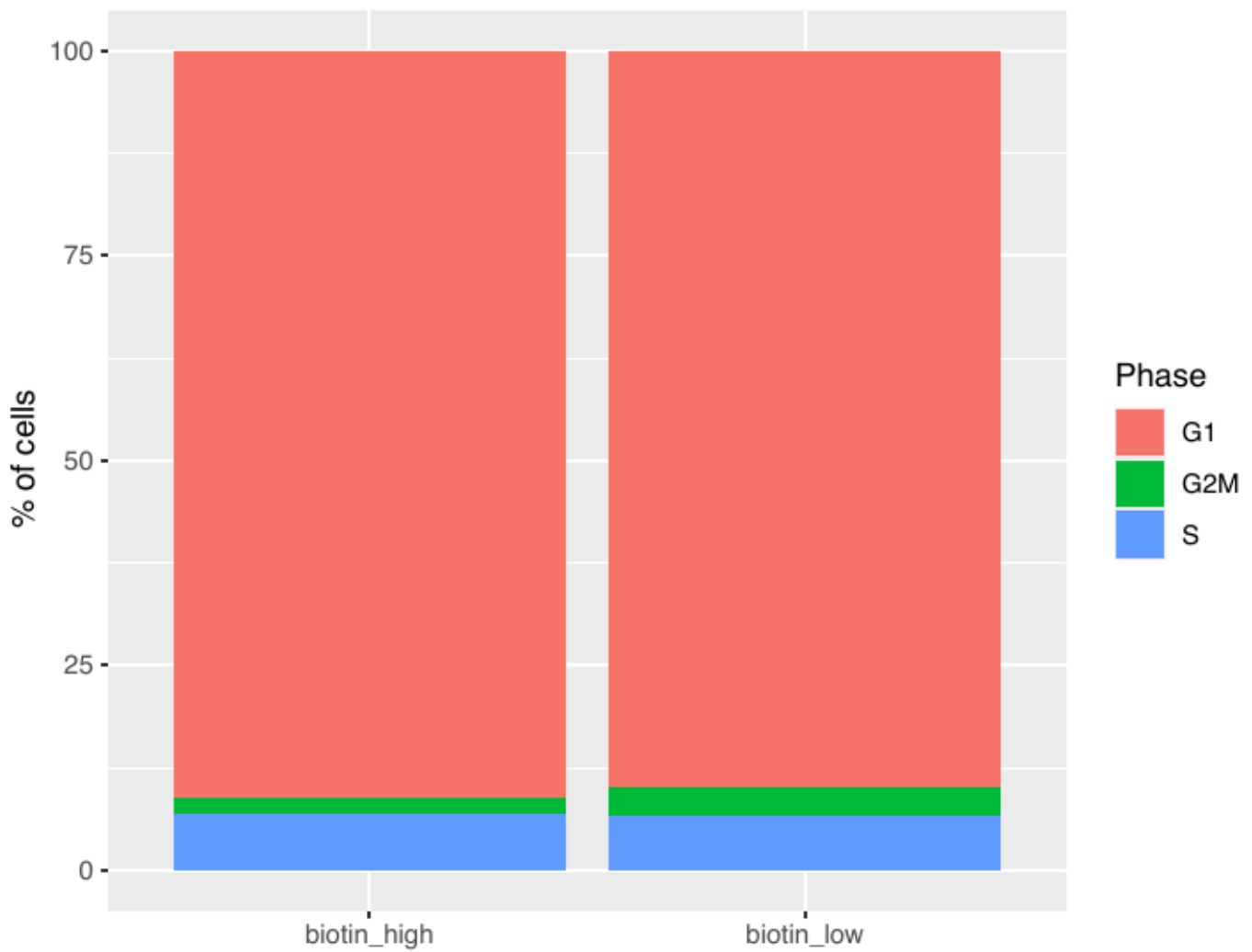


Cell cycle

4_cell_cycle_phase_proportions_biotin_high_vs_low.pdf

Cell-cycle phase proportions in the two groups, from Seurat CellCycleScoring.

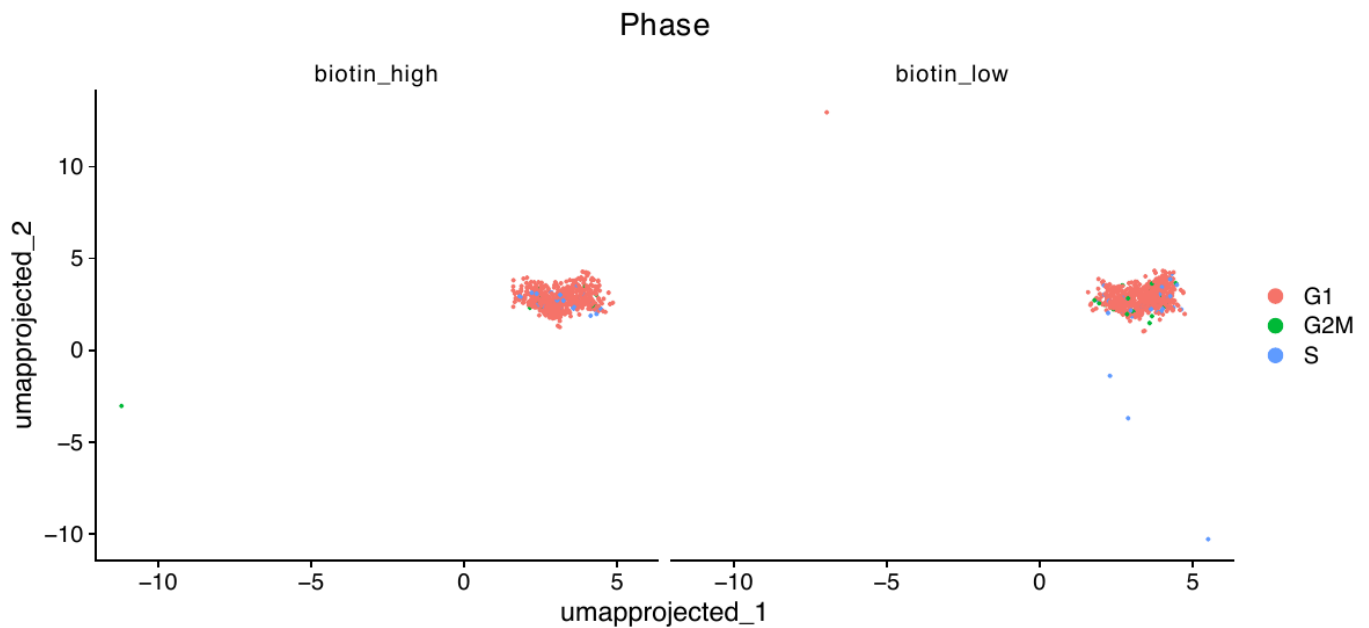
HSC MPP – cell-cycle phase proportions, biotin high vs low



5c_cell_cycle_phase_umap_projected.pdf

Phase on the projected UMAP, to check the split is not simply tracking proliferation.

HSC MPP – cell-cycle phase, biotin high vs low (Projected (BoneMarrowMap))



1_findmarkers_biotin_high_vs_low.csv

gene	p_val	avg_log2FC	pct.1	pct.2	p_val_adj
HLA-A	0.0000000	0.3779058	0.958	0.853	0.0000000
HLA-DRB1	0.0000000	0.4008393	0.918	0.813	0.0000004
CD74	0.0000000	0.3065561	0.981	0.936	0.0000004
HLA-DRA	0.0000000	0.3568816	0.915	0.812	0.0000014
JAML	0.0000000	0.7298979	0.421	0.264	0.0000218
HLA-B	0.0000000	0.3602259	0.950	0.889	0.0000225
ALOX5	0.0000000	0.8080985	0.387	0.238	0.0000264
CKLF	0.0000000	0.4570531	0.746	0.569	0.0000264
B2M	0.0000000	0.2477479	0.980	0.940	0.0001586
ITGB2	0.0000000	0.3951703	0.640	0.493	0.0003112
GPX1	0.0000000	0.3368304	0.931	0.863	0.0006495
CTSD	0.0000000	0.3665001	0.740	0.596	0.0007168
GLIPR2	0.0000000	0.4226815	0.462	0.320	0.0010304
HLA-DMA	0.0000000	0.3677610	0.711	0.571	0.0012998
CD36	0.0000001	0.5715461	0.420	0.282	0.0017373

1b_fgsea_hallmark_biotin_high_vs_low.csv

pathway	pval	padj	log2err	ES	NES	size	leadingEdge
HALLMARK_INTERFERON_ALPHA_RESPONSE	0.0000003	0.0000175	0.6749629	0.7067293	1.669594	48	CD74;B2M;HLA-C;PSMB8;IFITM3;SAMD9L;ITGB2
HALLMARK_INTERFERON_GAMMA_RESPONSE	0.0000044	0.0001097	0.6105269	0.6025433	1.474864	94	HLA-A;HLA-DRB1;CD74;HLA-B;B2M;HLA-DMA
HALLMARK_ALLOGRAFT_REJECTION	0.0000581	0.0009683	0.5573322	0.6189510	1.487961	63	HLA-A;CD74;HLA-DRA;B2M;ITGB2;HLA-DMA
HALLMARK_OXIDATIVE_PHOSPHORYLATION	0.0002711	0.0033892	0.4984931	0.5540784	1.361317	100	ATP1B1;SURF1;COX6A1;MRPL15;ATP6V1E1;ND6
HALLMARK_MYC_TARGETS_V1	0.0005450	0.0054496	0.4772708	0.5661953	1.383107	88	LDHA;DUT;PCBP1;DEK;PSMA1;HSP1;EIF4G1
HALLMARK_APICAL_JUNCTION	0.0012610	0.0105080	0.4550599	0.6034140	1.427791	50	ACTB;CD99;CAP1;GNAI2;VCL;GNAI1;ACTG1
HALLMARK_COMPLEMENT	0.0020534	0.0146672	0.4317077	0.5650914	1.361355	66	CTSD;CD36;CASP10;DPP4;TIMP1;PSMB9;CD3E
HALLMARK_COAGULATION	0.0033787	0.0211169	0.4317077	0.6476230	1.464987	26	DPP4;TIMP1;DUSP6;PEF1;CFD;RABIF;SERPINC1
HALLMARK_MYOGENESIS	0.0038528	0.0214042	0.4317077	0.6408595	1.465071	29	CD36;PTP4A3;SVIL;PDLIM7;MEF2C;LSP1;ATP2B4
HALLMARK_TNFA_SIGNALING_VIA_NFKB	0.0047220	0.0221396	0.4070179	0.5405460	1.308564	70	DUSP1;TNFAIP2;TSC22D1;ZFP36;MCL1;SATB1
HALLMARK_HYPOXIA	0.0048707	0.0221396	0.4070179	0.5625511	1.348408	61	DUSP1;SLC2A5;ZFP36;GAPDH;LDHA;DDIT4
HALLMARK_APOPTOSIS	0.0114335	0.0476398	0.3807304	0.5317281	1.288523	71	GPX1;IFITM3;TIMP1;MCL1;PMAIP1;SAT1;PTEN
HALLMARK_IL2_STAT5_SIGNALING	0.0249750	0.0938023	0.2878571	0.5109612	1.243654	78	GLIPR2;IFITM3;CTS2;IL3RA;CYFIP1;GSTO1;IL2
HALLMARK_ANDROGEN_RESPONSE	0.0280280	0.0938023	0.2712886	0.5537309	1.293734	40	B2M;TSC22D1;HERC3;SAT1;DBI;MYL12A;NFE2L3
HALLMARK_IL6_JAK_STAT3_SIGNALING	0.0281407	0.0938023	0.2712886	0.5855548	1.343808	30	CD36;IL3RA;PTPN1;IL10RB;MAP3K8;LTB;IL6

2_pseudobulk_deseq2_biotin_high_vs_low.csv

gene	logFC	logCPM	PValue	FDR	n_high	n_low
ENSG00000291215	-0.0226942	4.084848	0.9811972	0.999988	684	685
LINC01409	0.0235046	5.696121	0.9783209	0.999988	684	685
ENSG00000290784	1.0191247	3.425298	0.3443618	0.999988	684	685
LINC00115	0.2117707	3.513530	0.8378416	0.999988	684	685
LINC01128	0.5585734	4.977871	0.5348130	0.999988	684	685
NOC2L	0.0175761	7.175743	0.9831810	0.999988	684	685
ISG15	0.0508354	7.213841	0.9513551	0.999988	684	685
C1orf159	-0.0603082	5.685074	0.9444326	0.999988	684	685

gene	logFC	logCPM	PValue	FDR	n_high	n_low
TTLL10	0.4212241	4.959651	0.6395614	0.999988	684	685
TNFRSF18	0.4127231	6.858750	0.6226507	0.999988	684	685
TNFRSF4	0.1324141	8.644643	0.8721051	0.999988	684	685
SDF4	0.0094957	8.621808	0.9907897	0.999988	684	685
B3GALT6	0.0826427	6.351152	0.9223013	0.999988	684	685
UBE2J2	0.2865393	7.813023	0.7291838	0.999988	684	685
ACAP3	0.1996985	6.710629	0.8122232	0.999988	684	685

2_pseudobulk_edgeR_biotin_high_vs_low.csv

gene	logFC	logCPM	PValue	FDR	n_high	n_low
NFIA-AS2	-7.436254	0.9574418	0.0009007	1	684	685
ZNF502	-3.263622	1.4177032	0.0090302	1	684	685
ENSG00000223576	3.182053	1.1179739	0.0125866	1	684	685
AKAP12	3.735561	0.7733070	0.0128214	1	684	685
PLK2	-2.414068	2.5525250	0.0148245	1	684	685
CCDC80	3.648802	0.7073528	0.0162256	1	684	685
RCAN2	3.045422	1.0131098	0.0183347	1	684	685
BFSP2	-2.551518	1.4991026	0.0298844	1	684	685
HTR7	2.725889	0.7765933	0.0417569	1	684	685
ENSG00000260919	2.725889	0.7765933	0.0417569	1	684	685
HRH1	2.148100	1.4417468	0.0513877	1	684	685
CDKN1A	-1.856315	2.7518162	0.0562336	1	684	685
ENSG00000288942	2.155500	1.1735359	0.0578481	1	684	685
HDAC2-AS2	-1.961936	1.4936341	0.0601929	1	684	685
PEAK3	2.534953	0.6415567	0.0656334	1	684	685

3_cite_dsb_wilcoxon_biotin_high_vs_low.csv

protein	n_high	n_low	mean_high	mean_low	mean_diff	pvalue	FDR
biotin	684	685	2.0200481	0.2860675	1.7339806	0	0
CD99	684	685	4.9607297	3.3117641	1.6489655	0	0
EMR2	684	685	2.5437241	1.6374486	0.9062755	0	0
CD47	684	685	1.5874614	1.0654556	0.5220058	0	0
Fas	684	685	1.0367756	0.5651041	0.4716714	0	0
CD49e	684	685	0.8926937	0.5476195	0.3450742	0	0
CD33	684	685	1.5170085	1.0087878	0.5082207	0	0
CD45RA	684	685	1.8190673	1.2115961	0.6074712	0	0
CD18	684	685	1.0486318	0.4241690	0.6244628	0	0
LFA-3	684	685	0.5428249	0.2970006	0.2458244	0	0
HLA-ABC	684	685	1.9652520	1.3803815	0.5848705	0	0
CD54	684	685	1.2431237	0.7741792	0.4689444	0	0
CD49f	684	685	0.5854978	0.2894551	0.2960426	0	0
CD123	684	685	4.2958018	3.2325294	1.0632725	0	0
HLA-DR	684	685	1.2534349	0.8768929	0.3765420	0	0

3b_glyco_residual_vs_glycosyltransferase_RNA.csv

gene	coef	pvalue	FDR
------	------	--------	-----

gene	coef	pvalue	FDR
ST6GAL1	0.0272304	0.7139698	0.7356198
FUT8	0.0236438	0.7356198	0.7356198
MGAT5	0.0352814	0.4649772	0.7356198
B4GALT1	-0.0213735	0.6943826	0.7356198

3b_glyco_residual_vs_protein_residual.csv

protein	coef	se	pvalue	r_squared	FDR
CD18	0.2782090	0.0249493	0	0.0833771	0
CD99	0.1365042	0.0140432	0	0.0646498	0
CD49e	0.3985519	0.0432372	0	0.0585191	0
CD41	0.2411699	0.0273375	0	0.0538660	0
CD45RA	0.2344736	0.0291435	0	0.0452108	0
Fas	0.2689074	0.0342309	0	0.0431940	0
P-Selectin	0.2198772	0.0279963	0	0.0431740	0
EMR2	0.1660015	0.0216827	0	0.0411146	0
LFA-3	0.3355625	0.0452997	0	0.0385919	0
CD9	0.2327471	0.0314510	0	0.0385188	0
CD49f	0.2480743	0.0342590	0	0.0369402	0
CD33	0.2548260	0.0354929	0	0.0363380	0
CD61	0.1609890	0.0236225	0	0.0328597	0
CD47	0.2591014	0.0394148	0	0.0306434	0
CD36	0.1637732	0.0253101	0	0.0297185	0

4_cell_cycle_phase_proportions_biotin_high_vs_low.csv

biotin_group	Phase	n	pct
biotin_high	G1	624	91.228070
biotin_high	G2M	13	1.900585
biotin_high	S	47	6.871345
biotin_low	G1	616	89.927007
biotin_low	G2M	23	3.357664
biotin_low	S	46	6.715329

3b. The same split in monocytes

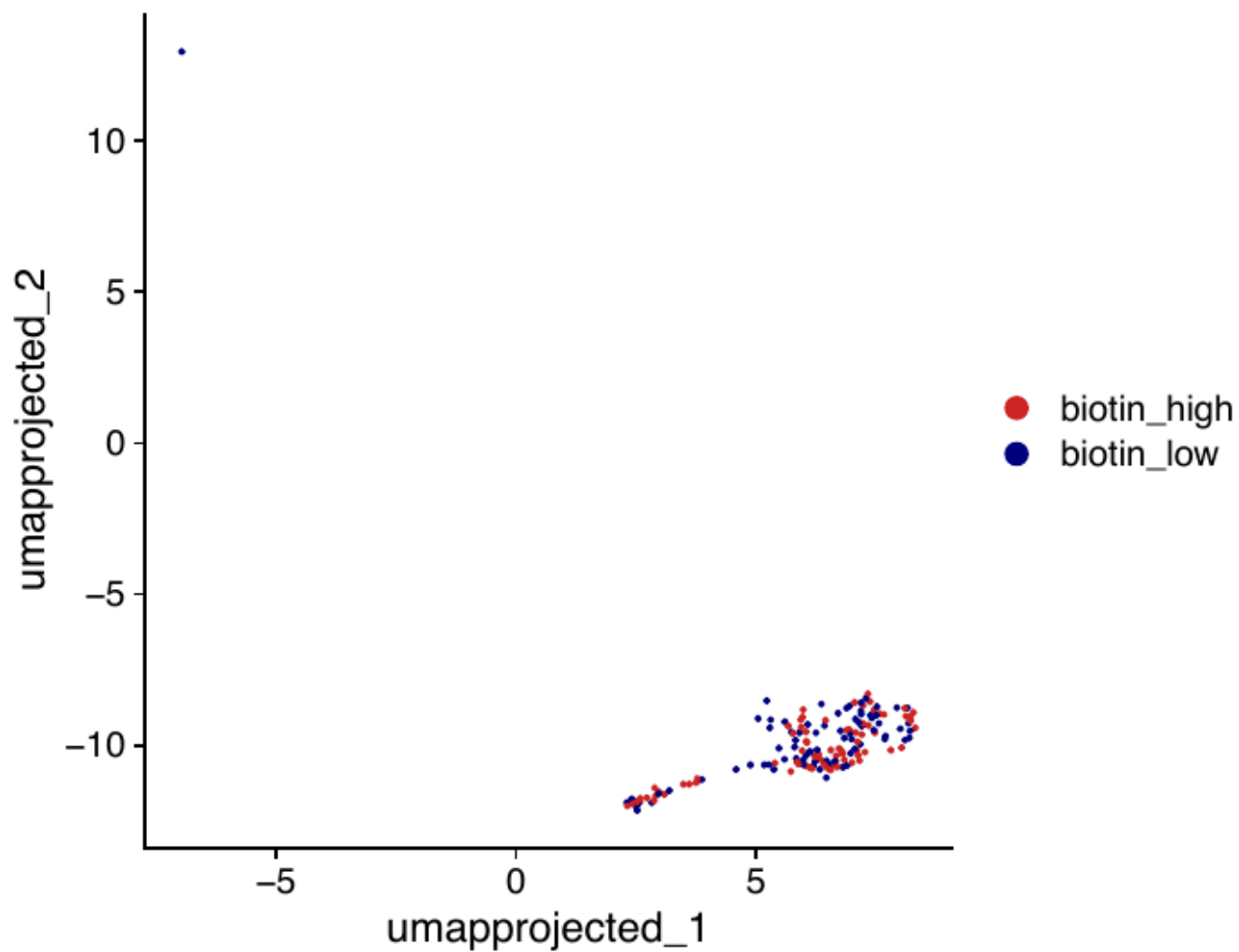
Monocytes carry the most biotin, so they are the obvious population to split. This arm had never been run - the report script existed but nothing wrote the object it needs. Step 02a is step 02 with the population taken as an argument; run as `Rscript 02a.biotin_split_analysis_runnable.R "Monocyte"` it produced everything below.

The population is small. 160 monocytes pass QC in this sample, splitting 80 high and 80 low at the median. That is a tenth of the HSC/MPP compartment, and it is why the pseudobulk contrast here is thinner - treat the gene-level results as indicative only.

5a_biotin_group_umap_projected.pdf

Biotin-high and biotin-low monocytes on the projected UMAP.

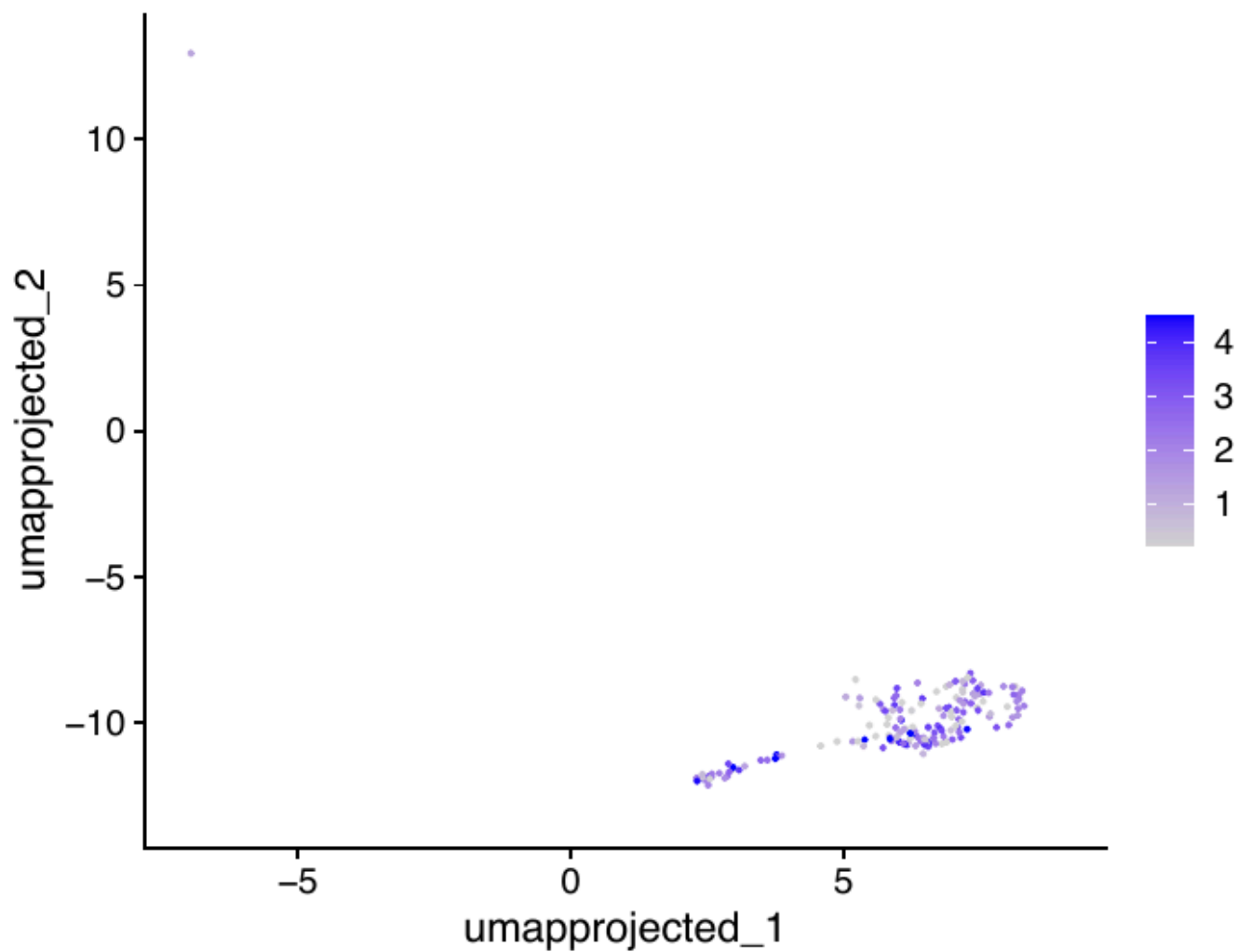
Monocyte – biotin high/low (BoneMarrowMap)



5b_biotin_value_umap_projected.pdf

Biotin as a continuous value across the same cells.

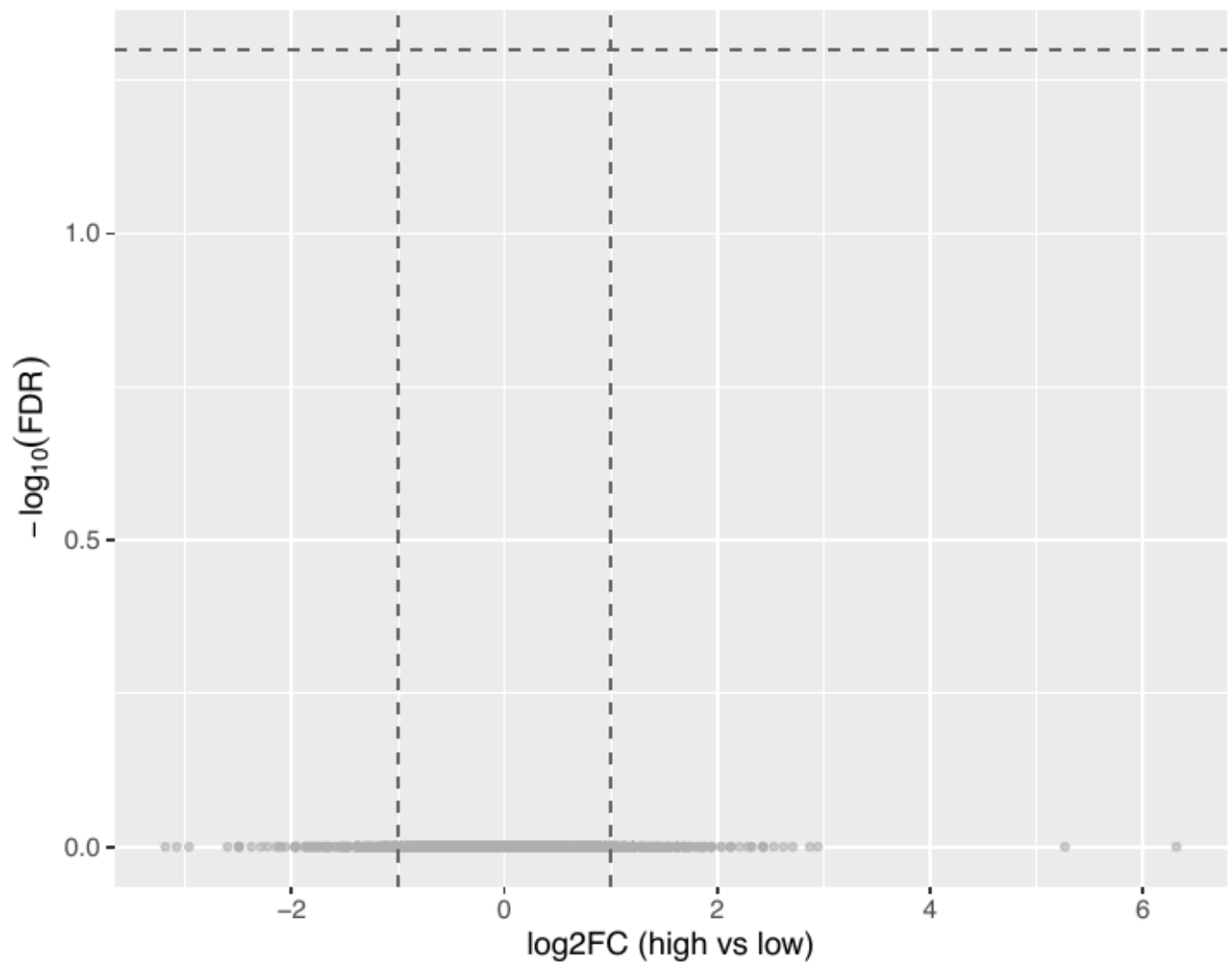
Monocyte – biotin value (BoneMarrowMap)



2_pseudobulk_deseq2_volcano_biotin_high_vs_low.pdf

Pseudobulk DESeq2, biotin-high against biotin-low monocytes.

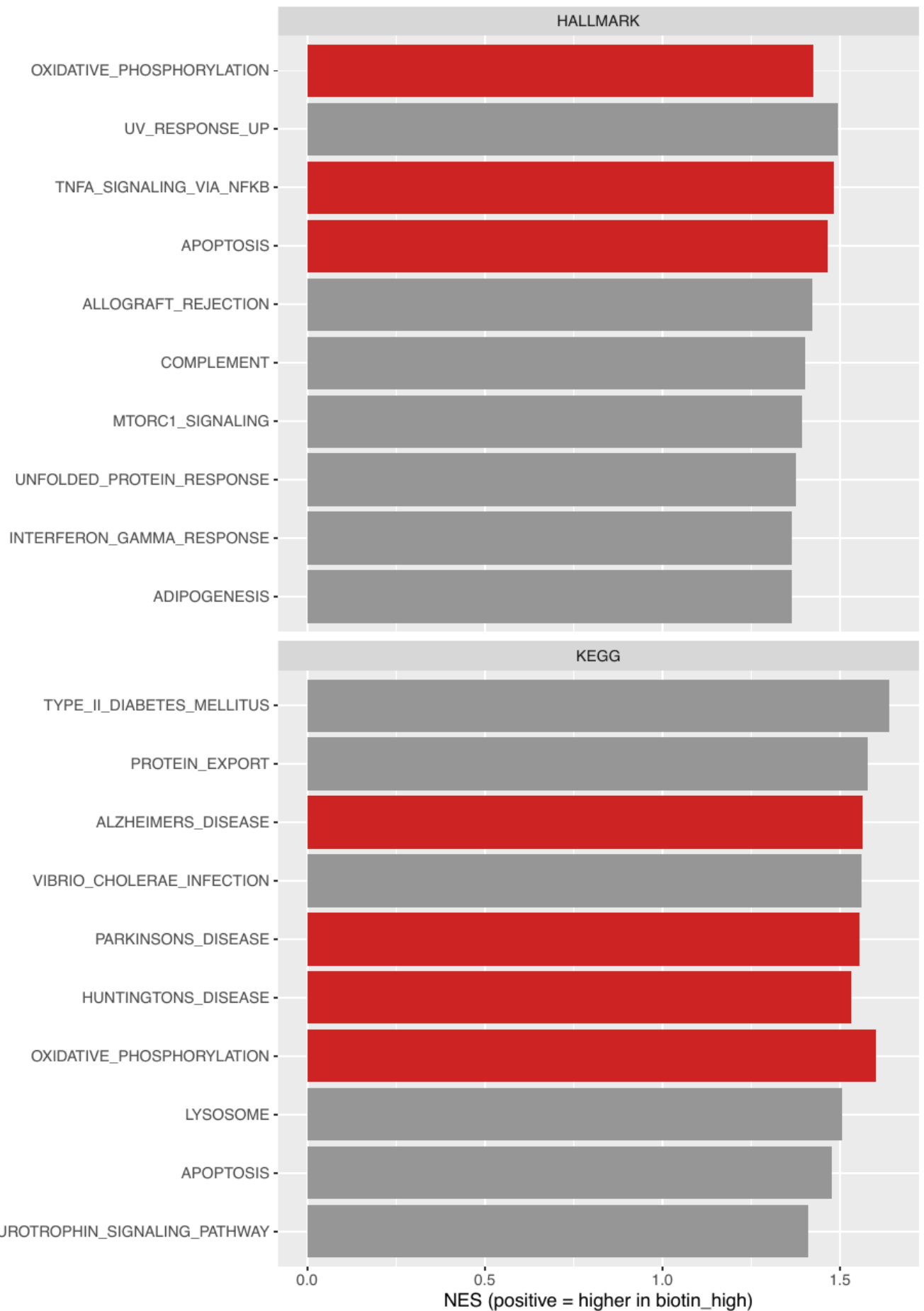
Monocyte – pseudobulk DESeq2 volcano, biotin high vs low



1b_fgsea_hallmark_kegg_biotin_high_vs_low.pdf

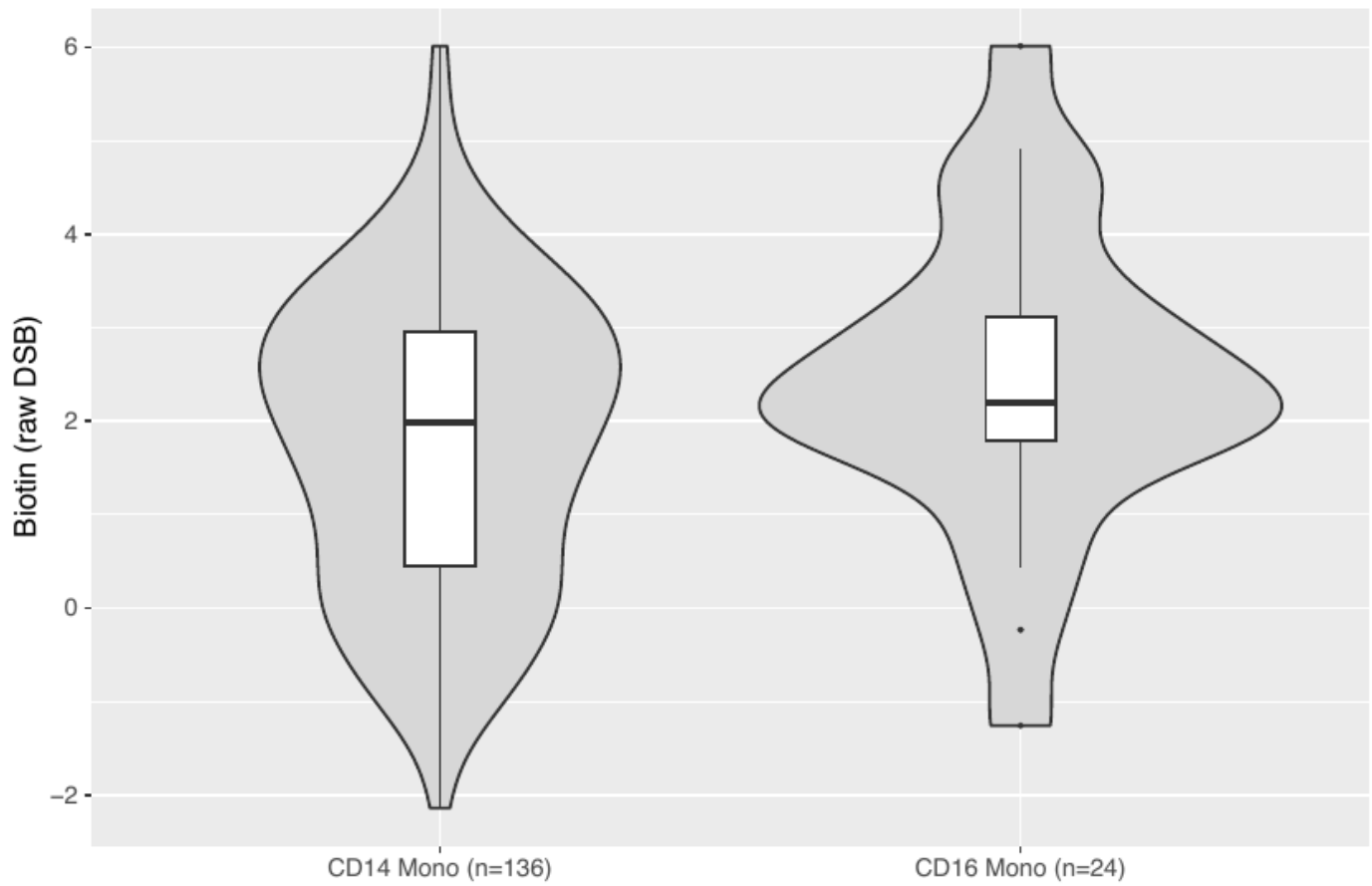
fgsea over Hallmark and KEGG on the FindMarkers ranking.

Monocyte – top Hallmark/KEGG pathways (fgsea), biotin high vs l



Glycosylation by fine cell type within the monocyte compartment.

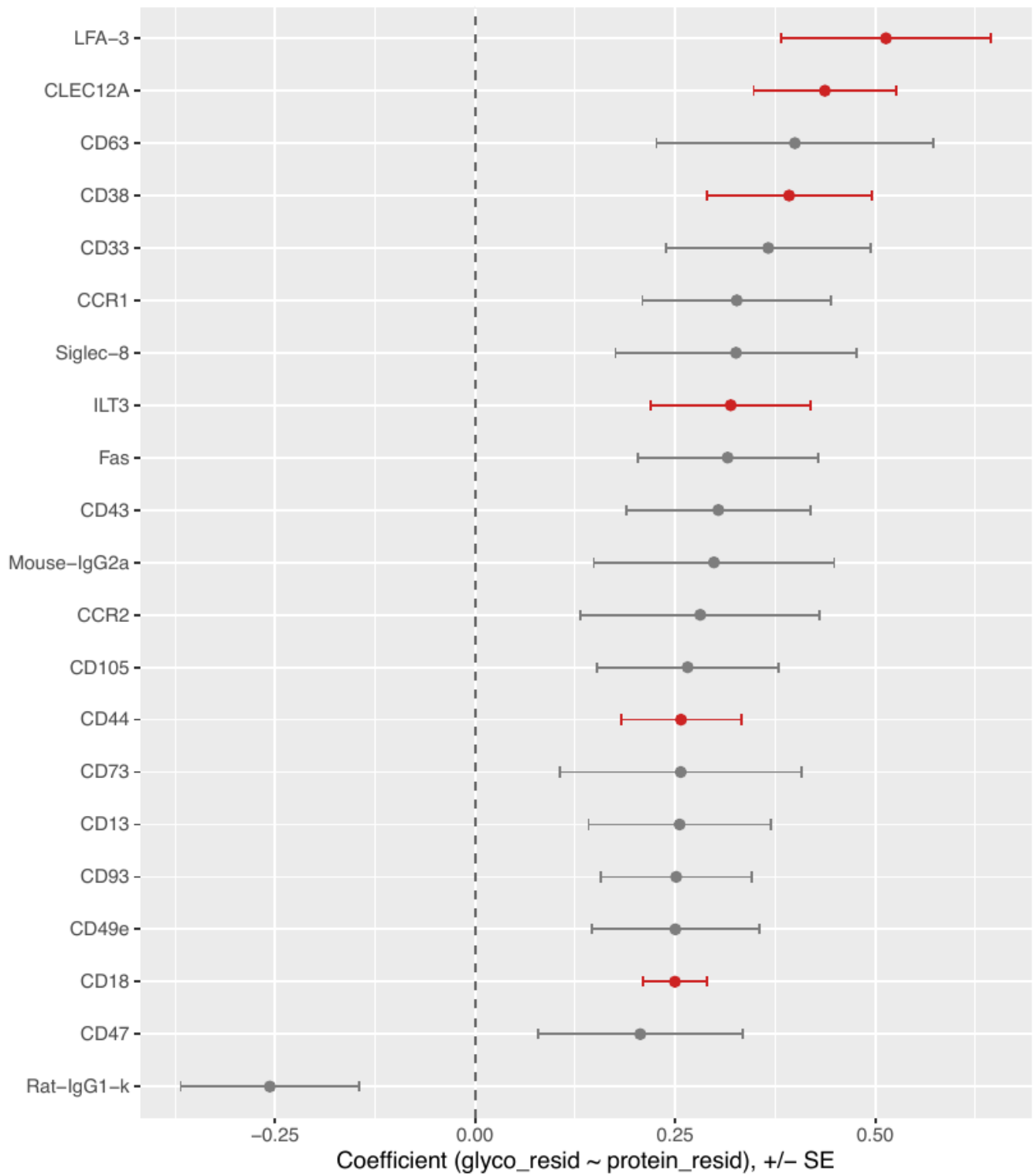
Monocyte (WEI21_26-17_NK) – biotin (glycosylation) by fine cell type



3c_glyco_residual_vs_protein_summary.pdf

Glycosylation against total surface load, as residuals.

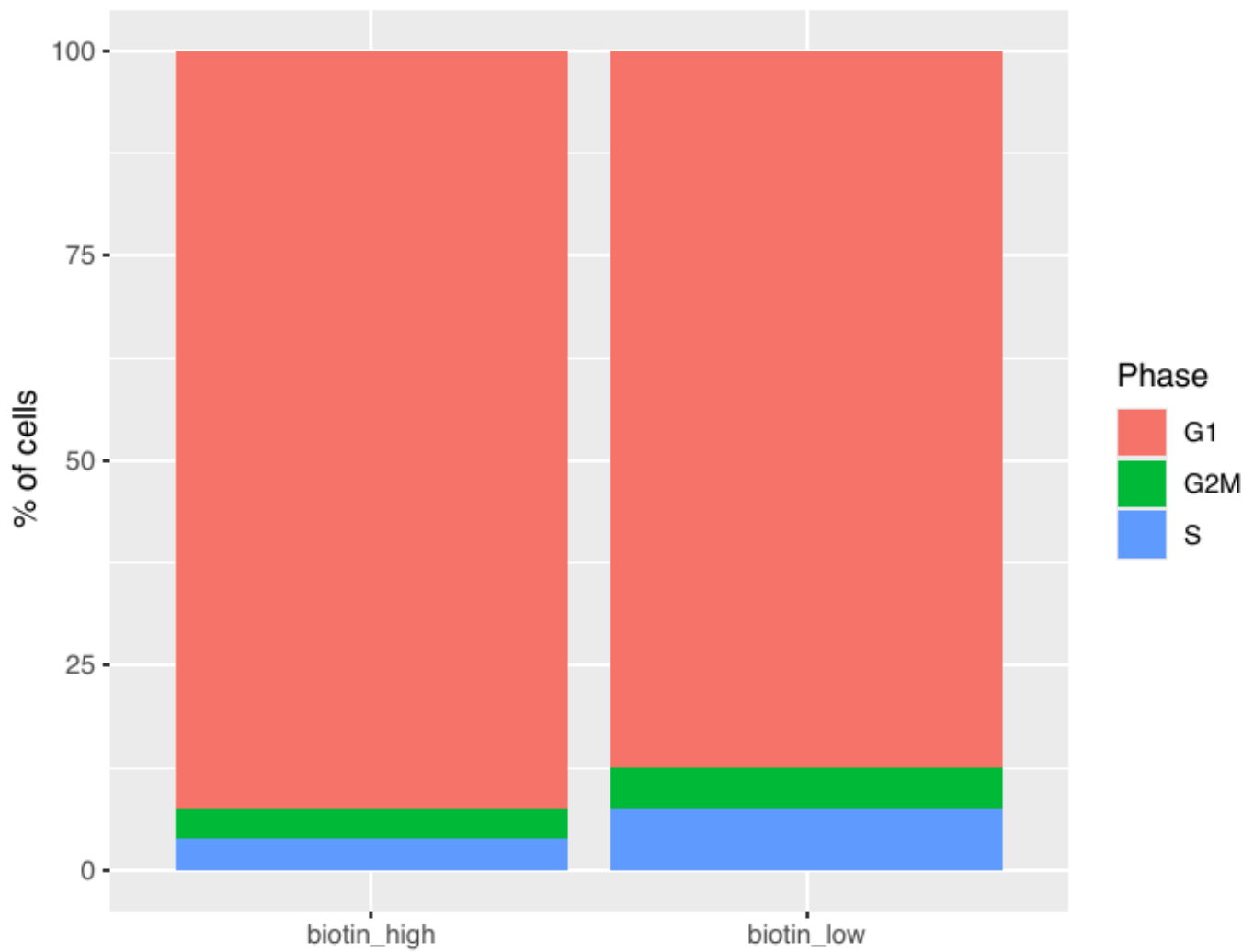
Monocyte (WEI21_26-17_NK) – proteins vs glycosylation residual



4_cell_cycle_phase_proportions_biotin_high_vs_low.pdf

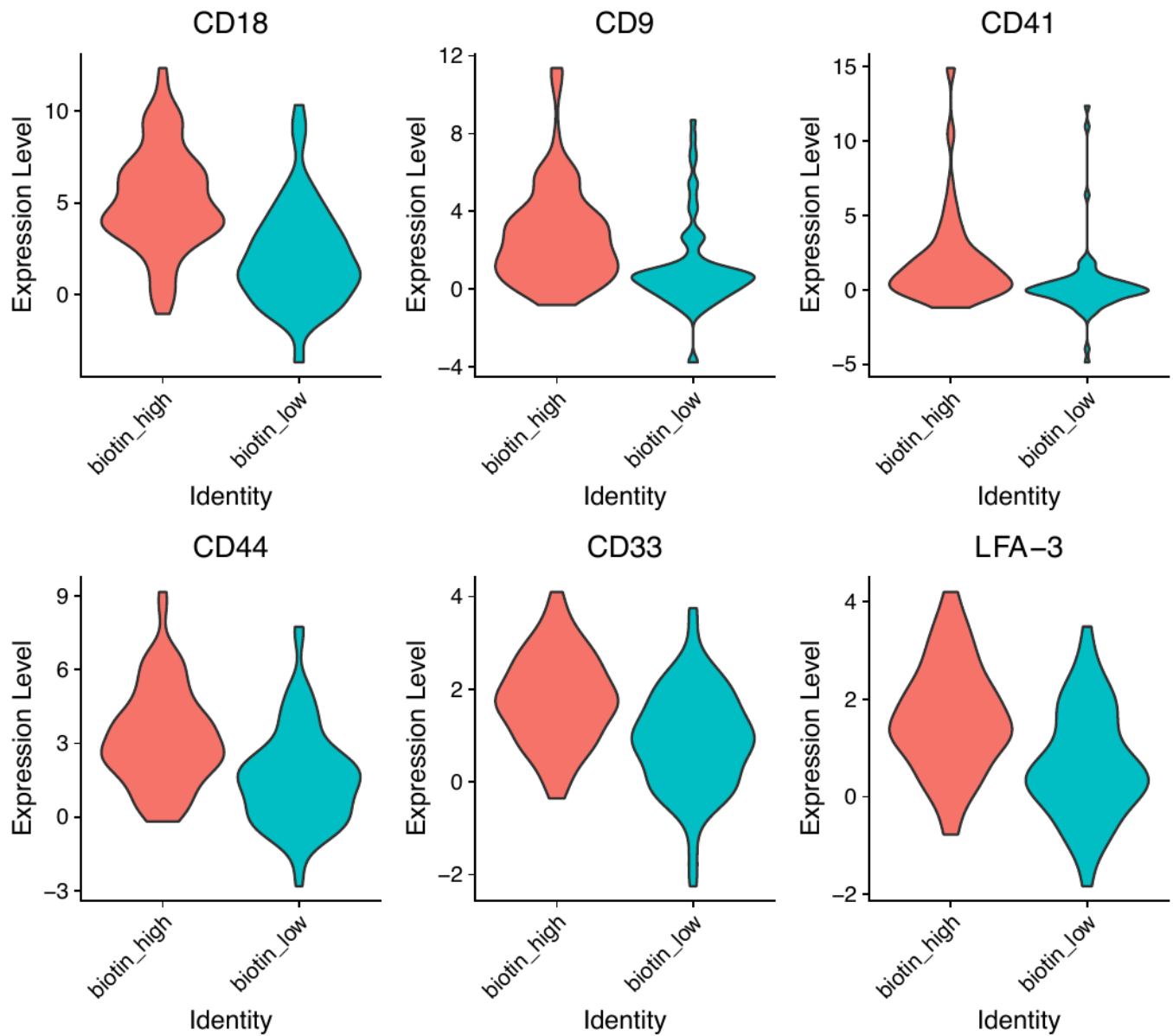
Cell-cycle phase by group. Both are overwhelmingly G1 - 92.5% of biotin-high and 87.5% of biotin-low - so the split is not separating cycling from non-cycling cells.

Monocyte – cell-cycle phase proportions, biotin high vs low



6c_cite_dsb_top_proteins_violin.pdf

Top differential CITE_DSB proteins between the groups.



3b_fine_celltype_counts.csv

predicted_CellType	n_cells
CD14 Mono	136
CD16 Mono	24

4_cell_cycle_phase_proportions_biotin_high_vs_low.csv

biotin_group	Phase	n	pct
biotin_high	G1	74	92.50
biotin_high	G2M	3	3.75
biotin_high	S	3	3.75
biotin_low	G1	70	87.50
biotin_low	G2M	4	5.00
biotin_low	S	6	7.50

3_cite_dsb_wilcoxon_biotin_high_vs_low.csv

protein	n_high	n_low	mean_high	mean_low	mean_diff	pvalue	FDR
biotin	80	80	3.168332	0.7621464	2.4061859	0.0000000	0.0000000

protein	n_high	n_low	mean_high	mean_low	mean_diff	pvalue	FDR
CD18	80	80	5.207457	2.3071295	2.9003277	0.0000000	0.0000000
CD9	80	80	2.811949	1.2337203	1.5782291	0.0000001	0.0000044
CD41	80	80	2.029657	0.6569892	1.3726678	0.0000001	0.0000051
CD44	80	80	3.253246	1.7457401	1.5075062	0.0000002	0.0000073
CD33	80	80	1.803108	1.0138290	0.7892788	0.0000004	0.0000122
LFA-3	80	80	1.645868	0.8188754	0.8269930	0.0000005	0.0000122
CLEC12A	80	80	1.986972	1.0879175	0.8990546	0.0000005	0.0000125
CD31	80	80	2.637690	1.4012343	1.2364553	0.0000007	0.0000153
CD45	80	80	2.326624	1.3186931	1.0079308	0.0000016	0.0000299
CD11b	80	80	3.052445	1.6126008	1.4398442	0.0000020	0.0000310
CD54	80	80	3.043211	1.7543823	1.2888290	0.0000020	0.0000310
P-Selectin	80	80	1.642489	0.6762009	0.9662876	0.0000025	0.0000362
EMR2	80	80	3.654302	2.2224542	1.4318474	0.0000028	0.0000371
CD11a	80	80	4.613571	2.8927332	1.7208373	0.0000030	0.0000371

Two differences from the HSC/MPP outputs are worth noting rather than glossing. This run writes its UMAPs into a `umaps/` subdirectory while the HSC/MPP figures sit flat, and it produces no edgeR volcano where HSC/MPP has one - both because the HSC/MPP outputs came from an earlier version of the analysis script than the one frozen here as step 02.

4. What was run, in order

Step	Script	What it does
01	<code>01.biotin_ADT_comparison.R</code>	Biotin ADT compared across the LK1 and LK2 runs.
02a	<code>02a.biotin_split_analysis_runnable.R</code>	Step 02 with the population as an argument, so any cell type can be split. Used to produce the monocyte arm.
02	<code>02.hsc_mpp_biotin_split_analysis.R</code>	HSC/MPP split on biotin, then RNA DE, fgsea, pseudobulk, CITE DE, glycosylation by cell type, surface-load residuals, cell cycle.
03	<code>03.biotin_split_report.Rmd</code>	The report over step 02's outputs.
04	<code>04.biotin_report.Rmd</code>	This report.

Three scripts are not kept as separate steps. `scripts/27b.hsc_mpp_biotin_split_report.Rmd` is the longer first version of step 03 and is superseded by it. `scripts/27d.monocyte_biotin_split_report_short.Rmd` is step 03 pointed at monocytes, differing by three lines - the title and two paths. It could not run before because nothing wrote the object it reads; step 02a now does, so it is reproducible. `scripts/27.removeme_quick_biotin_test.R` is scratch, named by its author as disposable.

5. Data locations

What	Location
Cross-run biotin comparison	<code>results/seurat_demux_test/LK1_vs_LK2_test_HT0_ADT_comparison/</code>
HSC/MPP split	<code>results/hsc_mpp_biotin_split/</code>
Monocyte split	<code>results/monocyte_biotin_split/</code> , figures under <code>umaps/</code>
Rendered prior reports	<code>scripts/27b.hsc_mpp_biotin_split_report.html</code> , <code>scripts/27c.hsc_mpp_biotin_split_report_short.html</code>
Slide figures	<code>figures_from_slides/</code> , from the 2026-07-22 progress deck

6. Tools

Tool	Version	Source
Seurat FindMarkers	see session below	Wilcoxon, single cell
DESeq2 / edgeR	see session below	pseudobulk, fixed dispersion

Tool	Version	Source
fgsea + msigdb	see session below	Hallmark gene sets
Seurat CellCycleScoring	see session below	phase assignment

```
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.5.2
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.1
##
## locale:
## [1] en_AU/en_AU/en_AU/C/en_AU/en_AU
##
## time zone: Australia/Melbourne
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] scales_1.4.0  ggplot2_4.0.2 knitr_1.51   purrr_1.2.1  stringr_1.6.0
## [6] readr_2.1.6   tidyr_1.3.2   dplyr_1.2.0
##
## loaded via a namespace (and not attached):
## [1] bit_4.6.0      gtable_0.3.6   jsonlite_2.0.0  crayon_1.5.3
## [5] qpdf_1.4.1     compiler_4.5.1 pdftools_3.9.0  Rcpp_1.1.1
## [9] tidyselect_1.2.1 parallel_4.5.1 jquerylib_0.1.4  yaml_2.3.12
## [13] fastmap_1.2.0  R6_2.6.1       generics_0.1.4  tibble_3.3.1
## [17] RColorBrewer_1.1-3 bslib_0.10.0   pillar_1.11.1   tzdb_0.5.0
## [21] rlang_1.1.7    cachem_1.1.0   stringi_1.8.7    xfun_0.56
## [25] S7_0.2.1       sass_0.4.10    bit64_4.6.0-1    ote1_0.2.0
## [29] cli_3.6.5      withr_3.0.2    magrittr_2.0.4   digest_0.6.39
## [33] grid_4.5.1     vroom_1.7.0    askpass_1.2.1    hms_1.1.4
## [37] lifecycle_1.0.5 vctrs_0.7.1    evaluate_1.0.5    glue_1.8.0
## [41] farver_2.1.2   rmarkdown_2.30 tools_4.5.1       pkgconfig_2.0.3
## [45] htmltools_0.5.9
```

7. Caveats

1. **The high/low split is one sample.** Everything in section 3 comes from WEI21_26-17_NK alone, so the RNA and protein differences are not replicated. Treat them as hypotheses.
2. **The monocyte arm rests on 160 cells**, 80 high and 80 low. It has now been run, but that is a tenth of the HSC/MPP compartment, so its gene-level results are weaker still.
3. **Both DESeq2 and edgeR volcanoes are shown** rather than one, because with a single sample the pseudobulk dispersion is fixed rather than estimated and the two methods can disagree.
4. **The cross-run comparison is LK1 against LK2 only.** LK3, which supplied the split sample, is not in it.